

Definition-Consistent Variable-Order Fractional Operators from Fixed-Pole Banks: Structure, Error Guarantees, and Order Tracking

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Abstract

Variable-order (VO) fractional operators have several inequivalent definitions that differ in how past orders enter the memory. Which definition a realization implements when it switches coefficients is unclear. An exact switching state map exists only between similar realizations, so realizations whose poles move with the order cannot implement the A- or B-type definitions under switching. A Beta-integral representation instead realizes the Grünwald–Letnikov weights with order-independent poles. With the order in the output weights the bank is the A-type, and with it in the input weights the B-type, for every order sequence; inverting it gives the recursive D- and E-types with exact duality, at $\mathcal{O}(K)$ cost. In the ℓ^1 and ℓ^∞ norms, the relative operator error of the forward types never exceeds the relative weight error; the recursive integrals amplify it by a computable condition number. The construction extends to every kernel family that is completely monotone in the lag, with $K = \mathcal{O}(\ln(1/\varepsilon) \ln(R/\varepsilon))$ common poles for tolerance ε and memory R , and explicit constants for Grünwald–Letnikov weights. A fit-free rule met all 400 specifications of a sealed holdout, and a DC floor makes every inverse stable. Moving-pole hot swapping stays at errors of 0.2 to 1 under switching. In online order tracking at 60 dB, filters on the bank reach median errors of 1.3×10^{-3} (A-type) and 1.2×10^{-2} (B-type), while the wrong definition leaves biases of 0.14 to 0.24. On measured lithium-ion drive cycles, running a fitted VO model with the other definition increased its held-out error in 51 of 56 cases.

Keywords: variable-order fractional calculus, Grünwald–Letnikov, fixed-pole approximation, completely monotone kernels, order tracking, Kalman filtering, lithium-ion battery, multifractional Brownian motion

1 Introduction

Fractional-order operators describe systems whose memory decays as a power law. When the decay exponent itself changes in time, the natural model is a variable-order (VO) operator, whose order $\alpha(t)$ is a function of time [1–4]. Unlike the constant-order case, the generalization is not unique. Different ways of letting the order enter the memory kernel give operators that coincide for constant order but differ as soon as the order changes [2, 5, 6].

For Grünwald–Letnikov (GL) operators, Sierociuk et al. use four types [7–10]:

- the A-type weights the whole history with the current order;
- the B-type weights each past sample with the order that was active when the sample entered;
- the recursive D- and E-types are the inverses of the A- and B-types of opposite order [8, 10, 11].

After a single order switch these definitions can differ by as much as the output itself (relative differences of 0.5 to 1.35 in Sect. 7). The choice of definition is therefore part of the model, not a detail of the numerics.

Practical realizations of fractional operators are finite-dimensional. Constant-order operators are approximated by rational transfer functions, most often by Oustaloup’s recursive distribution of poles and zeros [12], Charef’s fractional power pole [13] and related schemes [14]. The poles and zeros of these approximations move with the order. Fixed-pole approximations with order-independent poles have been proposed for constant orders [15–17]. Diffusive, infinite-state and sum-of-exponentials representations also lead to realizations with fixed nodes [18–22].

When the order varies, a constant-order realization can be switched between orders, either by switching circuits or by changing its coefficients while keeping its state (a “hot swap”). In our previous work on a constant-order FPGA core [23], such swaps produced short, bounded transients; which VO operator they implement was not addressed. Section 1.1 reviews how existing realizations handle a varying order.

This leaves two open questions:

1. Given an arbitrary order sequence, which VO definition does a finite-dimensional realization implement?
2. Can the structure of the realization select the definition at no extra cost per sample?

This paper answers both for GL operators of order $-1 < \alpha < 1$ and, more generally, for kernel families that are completely monotone in the lag. The contributions are:

1. *Necessity.* An exact switching state map exists only between similar realizations (Theorem 6). Realizations with order-dependent poles therefore cannot implement the A- or B-type exactly at a switch, with or without a state map. Least-squares maps reduce the error in floating point, but they are dense, depend on the input class, and lose three to nine orders of magnitude of accuracy in 24-bit fixed point (Sect. 7.3).
2. *Definition by structure.* Order-independent exponents have already been used for the A-type VO Caputo derivative in fast numerical solvers [24, 25]. We show that, in

a fixed-pole structure, the place of the order selects the definition. A Beta-integral representation of the GL weights gives a discrete bank with order-independent poles. With the order in the output weights it realizes the A-type, and with the order in the input weights the B-type, for every order sequence up to the frozen-order quadrature error (Proposition 2). Inverting the same bank gives the D- and E-types with exact duality at the same $O(K)$ cost per sample (Proposition 4); in analog form, a direct realization of the D-type has been considered impractical [10].

3. *Operator-level guarantees.* The weight error bounds the realized operators as a whole, for every order sequence (Theorem 5). In the ℓ^1 and ℓ^∞ operator norms the relative error of the A- and B-types never exceeds the relative weight error. The recursive integrals amplify it by the condition number of the inverted derivative, which grows like $N^{|\alpha|}$; one step response of the bank certifies their error a posteriori.
4. *Completely monotone families.* The structural results hold for every kernel family that is completely monotone in the lag, including time-varying distributed orders and fixed tempering. Under analyticity and power-law conditions on the density, a common pole set of size $K = O(\ln(1/\varepsilon) \ln(R/\varepsilon))$ meets a relative tolerance ε over a memory R uniformly in the order (Theorem 9). For the GL weights all constants are explicit (Lemma 10), which gives a fully a-priori design.
5. *Design and stability.* An analytic error model with three closed-form terms sets the bank size from the tolerance, the memory length and the order range; it met every tolerance of a sealed holdout of 400 random specifications (Sects. 5 and 7.4). The inverse of a derivative bank is stable if and only if its realized DC gain is positive (Proposition 8). This is tested through an exact sign and guaranteed by a DC floor, also after quantization. Against moving-pole hot swapping, truncated convolution and fixed poles with residues fitted by linear programming, only the fixed-pole structure reaches small errors under switching; fitted residues reduce the bank by 20 to 50 %, but keep the stability certificate only with sign constraints (Sect. 7.8).
6. *Applications.* In online tracking of a time-varying order, the A-type bank gives regressors that do not depend on the order, so an extended Kalman filter costs $O(K)$ and an exact grid filter $O(GK)$ per sample; the B-type bank gives a finite state for an augmented filter. Assuming the wrong definition biases the estimate by 0.14 to 0.24 at high SNR. Against unscented filters on the direct GL sum, the bank gives the full-memory accuracy at $O(K)$ instead of $O(n)$ operations per sample, and a truncated GL memory biases the estimate (Sect. 8.1). On measured drive cycles of a lithium-ion cell, running a fitted VO model with the operator of the other type increased its held-out error in 51 of 56 cases, by a median factor of 4.4 for B-type models (Sect. 8.2). The bank also streams multifractional Brownian motion of both types (Sect. 8.3), and a fixed-point core, bit-exact on an FPGA board, changes order and type at every sample through a ROM address (Sect. 9).

Section 1.1 reviews related work. Section 2 recalls the definitions and their duality. Section 3 builds the continuous- and discrete-time fixed-pole banks, Section 4 proves the structural results and the operator-norm bounds, and Section 5 gives the error model, the design rule and the stability result. Section 6 extends the construction to completely monotone kernel families. Section 7 reports the numerical experiments, and Section 8 three applications: order tracking, identification on measured battery

data and multifractional Brownian motion. Section 9 summarizes the fixed-point core, and Section 10 discusses limitations. Secondary results and implementation details are in the Supplementary Material. All code, seeds and results are available (see the Declarations).

1.1 Related work

The search behind this section is documented in the repository: protocol, queries, screening decisions and data extraction. It used Crossref and arXiv and was complemented by snowballing. Table 1 summarizes how existing realizations treat a varying order.

Analog realizations.

Passive domino and nested ladders already show variable-order behaviour in the time domain [26]. Most analog VO realizations, however, switch between two or three constant-order circuits [27, 28], as noted in [29, 30]. Sierociuk et al. made the link between structure and definition explicit for switching:

- a serial cascade in which a block of complementary order is pre-connected at every switch is equivalent to the B-type [7];
- the recursive types are dual to the iterative ones [8, 31], and switching strategies for them follow from this duality [8, 9, 32];
- a parallel scheme replaced the serial one, which could not be built in practice [33].

The D-type was found impractical to realize directly, because two ideal operators of opposite orders cannot be connected in series with electronic elements [10]. The serial scheme adds a block at every switch [7], and the analog models built so far cover a few switches. Arıcioglu realized a smoothly varying order with a single transfer function [29]. This restricts the method to the only linear time-invariant definition of Lorenzo and Hartley [2]. Electronically tunable fractional-order elements and filters [34, 35] and variable fractional-order digital differentiators and integrators [36–38] make the order an adjustable parameter of a constant-order design. Yu et al. went further and changed the element values of a fractal-ladder fractor with programmable resistors and capacitors, both in jumps and continuously [30]. They compared the response with a GL reference whose weights take the order at the time of each sample, i.e. a B-type reference. Because the element values, and hence the poles, change with the order, Theorem 6 below implies that such a match can only be approximate.

Digital and hardware realizations.

Tolba et al. realized GL operators on an FPGA with an order that can be reconfigured at run time, and used them in a VO chaotic oscillator [39]. A direct GL convolution whose weights are recomputed for the current order has fixed poles (all at the origin) and its order in the output weights. By the argument of Proposition 2 below it is then an A-type operator, with a memory and a cost per sample proportional to its length. The discrete-time VO PID controller of Oziablo et al. evaluates exactly such an A-type GL difference [40]. Its cost per step grows with time, and the authors note that bounding the stored history without disturbing the control is an open problem.

Fixed nodes for variable orders.

The work closest to ours comes from fast solvers for VO time-fractional equations. Zhang, Fang and Sun approximate the VO Caputo derivative, with the order taken at the current time, by an exponential sum whose exponents do not change between time levels [24, 25]. Huang et al. propose sum-of-exponentials approximations for variable-order kernels within a unified fast time-stepping framework [41]. The exponential-sum approximation of Zhang et al. shares the order-independent nodes with the present paper. However, it treats only the A-type, in continuous time and inside an L1-type discretization. It does not address the B-, D- or E-types, a design rule over the whole order range, or hardware. For the hidden-memory VO derivative, whose order is evaluated at the integration variable as in the B-type, the fast method we found uses Taylor expansions and hierarchical matrices at $O(N \log N)$ total cost [42].

Time-varying fractional filters.

In time-series analysis, Philippe, Surgailis and Viano introduced two time-varying fractional filters $A(d)$ and $B(d)$ built from a sequence d_t [43, 44]. Both reduce to $(I - L)^{-d}$ for constant d and satisfy $B(-d)A(d) = A(-d)B(d) = I$. Their coefficients are products of d_u over the times between input and output, so they differ from the GL types used here. Their exact inverse relation is, however, an analogue of the duality (7). Surgailis extended this construction to continuous time to define multifractional processes [45].

State and transients.

Transients caused by coefficient changes in recursive filters are a classical topic [46, 47]. The meaning of the state of a fractional system has been discussed in the context of initialization [19, 48, 49]. Neither line asks which VO definition a switched realization implements.

2 Variable-order Grünwald–Letnikov operators

2.1 Definitions

Let T_s be the sampling period and $\alpha_n \in (-1, 1)$ the order at sample $n \geq 0$. An order $\alpha > 0$ is a derivative, $\alpha < 0$ an integral and $\alpha = 0$ the identity. Signals vanish for $n < 0$. We write

$$w_r(a) = (-1)^r \binom{a}{r} = w_{r-1}(a) \frac{r-1-a}{r}, \quad w_0 = 1, \quad g_r(a) = T_s^{-a} w_r(a). \quad (1)$$

Definition 1 (VO GL operators [7, 8, 10]) For an input x and an order sequence α ,

$$({}^A\Delta^\alpha x)_n = \sum_{r=0}^n g_r(\alpha_n) x_{n-r}, \quad (2)$$

$$({}^B\Delta^\alpha x)_n = \sum_{r=0}^n g_r(\alpha_{n-r}) x_{n-r}, \quad (3)$$

Table 1 How existing realizations treat a varying order. “Inferred” marks entries obtained by applying Proposition 2 or Theorem 6 to the structure described in the cited work

Work	Structure	Definition	Order changes	Cost per sample
serial switching [7]	analog; complementary-order block added at each switch	B	any number of switches	grows with switches
recursive, parallel schemes [8, 10, 33] [29]	analog; switched constant-order ladders analog; one transfer function	D; B; B $\leftrightarrow$ D LTI (third) type of [2]	a few switches smooth $\alpha(t)$	— fixed
tunable elements, filters [34–36, 38]	order as a design parameter	FIR with current-order coefficients: A (inferred)	re-tuning	—
programmable ladder [30]	element values change with the order (moving poles)	compared with B; approximate (Thm. 6)	jumps, continuous	fixed
FPGA GL [39]; VO PID [40]	direct GL convolution	A (inferred [39]; stated [40])	at run time	$O(L)$, $L =$ memory
hot-swapped Oustaloup (Sect. 7)	moving poles, state kept	none exactly (Thm. 6)	any	$O(S)$
exponential sums [24, 25]	fixed exponents, continuous-time Caputo	A	any $\alpha(t) \in (0, 1)$	$O(\log^2 n)$
hidden memory [42]	Taylor / hierarchical matrices	B-like (Caputo)	any	$O(N \log N)$ in total
this work	discrete fixed-pole bank	A, B; D, E by inversion	every sample	$O(K)$, $K \approx 20\text{--}50$

$$z_n = \left({}^D\Delta^\alpha x \right)_n = T_s^{-\alpha_n} x_n - \sum_{r=1}^n w_r(-\alpha_n) z_{n-r}, \quad (4)$$

$$z_n = \left({}^E\Delta^\alpha x \right)_n = T_s^{-\alpha_n} x_n - \sum_{r=1}^n w_r(-\alpha_{n-r}) T_s^{\alpha_{n-r}-\alpha_n} z_{n-r}. \quad (5)$$

The A-type has no memory of past orders: a change of α_n immediately reweights the entire history. The B-type keeps each sample with the weight sequence of its own entry order. For a constant order all four types coincide. In [7] the A- and B-types are the first and second types. Their third type indexes the order by the lag, which gives a time-invariant operator. Lorenzo and Hartley [2] and Valério and Sá da Costa [5] list further variants, and [50] exploits the duality below to solve VO equations analytically. The time-varying fractional filters of [43, 44] form a different family with an exact inverse relation of the same kind.

2.2 Matrix form and duality

On a horizon of N samples the A- and B-types are lower-triangular matrices,

$$[W_A(\alpha)]_{n,j} = g_{n-j}(\alpha_n), \quad [W_B(\alpha)]_{n,j} = g_{n-j}(\alpha_j), \quad 0 \leq j \leq n < N. \quad (6)$$

Row n of $W_A(-\alpha)z = x$ reads $T_s^{\alpha_n} z_n + T_s^{\alpha_n} \sum_{r \geq 1} w_r(-\alpha_n) z_{n-r} = x_n$, which is (4). Row n of $W_B(-\alpha)z = x$ is (5). Hence

$$W_D(\alpha) = W_A(-\alpha)^{-1}, \quad W_E(\alpha) = W_B(-\alpha)^{-1}, \quad (7)$$

which is the duality of [8, 10, 11]. For a varying order, the A-types (or the B-types) of opposite orders are not inverses of each other: $W_A(\alpha)W_A(-\alpha) \neq I$ in general. The recursive types are therefore genuinely different operators. Section 7.5 quantifies these relations numerically.

2.3 Continuous-time references

For the continuous-time experiments we use the step responses of the A- and B-types. With $F(t, a) = t^{-a}/\Gamma(1-a)$, the step response of the constant-order operator s^a , they are

$$y_A(t) = F(t, \alpha(t)), \quad (8)$$

$$y_B(t) = F(t, \alpha(0)) + \int_0^t \partial_a F(t - \tau, \alpha(\tau)) \dot{\alpha}(\tau) d\tau. \quad (9)$$

Equation (9) is derived in Supplementary Sect. S6. For a piecewise-constant order with switches at T_i , (9) reduces to $F(t, \alpha_0) + \sum_i [F(t - T_i, \alpha_i) - F(t - T_i, \alpha_{i-1})]$.

2.4 Diffusive representation

For $0 < a < 1$ and $S_a = \sin(\pi a)/\pi$ [18, 22],

$$s^{-a} = S_a \int_0^\infty \frac{\xi^{-a}}{s + \xi} d\xi, \quad s^a = S_a \int_0^\infty \xi^{a-1} \frac{s}{s + \xi} d\xi. \quad (10)$$

The order enters only through the density in ξ ; the pole $-\xi$ of each element does not depend on it. This is the property the paper builds on.

3 Fixed-pole banks

3.1 Continuous time

Let $\xi_k = \xi_{\text{lo}} e^{kh}$, $k = 0, \dots, K-1$, be a geometric grid with spacing h in $\ln \xi$, and apply the trapezoidal rule to (10). Every order is then written on the same bank of unit-DC-gain states $\dot{x}_k = -\xi_k x_k + \xi_k u$:

$$H_\alpha(s) = d(\alpha) + \sum_{k=0}^{K-1} m_k(\alpha) \frac{\xi_k}{s + \xi_k}. \quad (11)$$

For an integral, $\alpha = -a < 0$, the weights are $m_k = h S_a \xi_k^{-a}$. For a derivative, $\alpha = a > 0$, we use $s/(s + \xi) = 1 - \xi/(s + \xi)$: then $m_k = -v_k$ with $v_k = h S_a \xi_k^a$, and the

feedthrough collects $\sum_k v_k$. The two truncated tails are closed by the geometric series of the virtual trapezoid nodes beyond the grid:

- one tail is lumped onto the outermost pole: $m_0 += S_a \xi_{lo}^{-a} G_h(1-a)$ for integrals, $v_{K-1} += S_a \xi_{hi}^a G_h(1-a)$ for derivatives;
- the other becomes feedthrough: $d = S_a \xi_{hi}^{-a} G_h(a)$ for integrals, $d = \sum_k v_k + S_a \xi_{lo}^a G_h(a)$ for derivatives.

Here

$$G_h(p) = \sum_{j \geq 1} h e^{-pjh} = \frac{h}{e^{ph} - 1}, \quad H_h(p, q) = G_h(p) - G_h(q). \quad (12)$$

All nodes carry the full weight h . The bank therefore equals the infinite trapezoidal rule except for asymptotic tail residuals, and no $O(h^2)$ endpoint term appears. At $\alpha = 0$ the bank is the identity ($m = 0$, $d = 1$). Since $S_a G_h(a) \rightarrow 1$ as $a \rightarrow 0$, the coefficients are continuous when the order crosses zero.

3.2 Discrete time: a Beta-integral representation of the GL weights

Sampling the continuous-time bank introduces a second error, because modes faster than $1/T_s$ respond to the staircase of a sampled order (Sect. 7.2). We therefore realize the GL weights (1) themselves.

Lemma 1 For $r \geq 1$ and $-1 < a < 1$,

$$w_r(a) = -S_a B(r-a, 1+a) = -S_a \int_0^1 \theta^{r-a-1} (1-\theta)^a d\theta. \quad (13)$$

Proof $w_r(a) = \Gamma(r-a)/(\Gamma(-a)\Gamma(r+1))$ and $B(r-a, 1+a) = \Gamma(r-a)\Gamma(1+a)/\Gamma(r+1)$. The reflection formula gives $\Gamma(-a)\Gamma(1+a) = -\pi/\sin(\pi a)$. For $a = 0$ both sides vanish. $\square$

The weights are thus a superposition of geometric sequences θ^r with an order-dependent density. With $\theta = e^{-u}$ and $u = e^x$,

$$w_r(a) = -S_a \int_{-\infty}^{\infty} \varphi_a(e^x) e^{-(r-1)e^x} dx, \quad \varphi_a(u) = u e^{-(1-a)u} (1 - e^{-u})^a. \quad (14)$$

The trapezoidal rule on $u_k = u_{lo} e^{kh}$, $k = 0, \dots, K-1$, gives order-independent discrete poles $\theta_k = e^{-u_k}$. With $u_k = \xi_k T_s$, each θ_k is the zero-order-hold image of the continuous-time pole $-\xi_k$. The realized weights are

$$\begin{aligned} \hat{g}_0(a) &= T_s^{-a}, & \hat{g}_r(a) &= \delta_{r1} c_d(a) + \sum_{k=0}^{K-1} c_k(a) \theta_k^{r-1} \quad (r \geq 1), \\ c_k(a) &= -S_a T_s^{-a} h \varphi_a(u_k), \end{aligned} \quad (15)$$

and the frozen-order transfer function is

$$\hat{H}_a(z) = \sum_{r \geq 0} \hat{g}_r(a) z^{-r} = T_s^{-a} + \frac{c_d(a)}{z} + \sum_{k=0}^{K-1} \frac{c_k(a)}{z - \theta_k}. \quad (16)$$

The tails are again closed by virtual trapezoid nodes:

- the nodes below the grid ($\theta \approx 1$) are lumped onto θ_0 , matched at $r = 1$: $c_0 += -S_a T_s^{-a} \sum_{j \geq 1} h \varphi_a(u_{\text{lo}} e^{-jh})$;
- the nodes above the grid ($\theta \approx 0$) contribute only at $r = 1$ and form the *delay mode* (a pole at $\theta = 0$): $c_d = -S_a T_s^{-a} \sum_{j \geq 1} h \varphi_a(u_{\text{hi}} e^{jh})$.

Since $\varphi_a > 0$, all residues have the sign of $-a$: integral banks have positive residues and derivative banks negative ones (before the DC floor of Sect. 5.4). The weight $\hat{g}_0$ is exact, and at $a = 0$ the bank is exactly the identity. As $a \rightarrow 0$ all residues vanish continuously, so the bank crosses the order zero without any special case.

3.3 Order schedules

The bank has K first-order states s_k and the delay state s_d . The order can be scheduled in two places (Fig. 1), the output schedule (OS) and the input schedule (IS):

$$\begin{aligned} \text{OS:} \quad & s_k[n+1] = \theta_k s_k[n] + v_n, \quad s_d[n+1] = v_n, \\ & y_n = T_s^{-\alpha_n} v_n + \underbrace{c_d(\alpha_n) s_d[n] + \sum_k c_k(\alpha_n) s_k[n]}_{\eta_n}, \end{aligned} \quad (17)$$

$$\begin{aligned} \text{IS:} \quad & s_k[n+1] = \theta_k s_k[n] + c_k(\alpha_n) v_n, \quad s_d[n+1] = c_d(\alpha_n) v_n, \\ & y_n = T_s^{-\alpha_n} v_n + \underbrace{s_d[n] + \sum_k s_k[n]}_{\eta_n}. \end{aligned} \quad (18)$$

In both cases $y_n = \hat{g}_0(\alpha_n) v_n + \eta_n$, where the history η_n depends only on $v_0, \dots, v_{n-1}$ because the poles do not depend on the order. Changing the order never touches the state.

4 Definition by structure

Proposition 2 (Definition by structure) *For every order sequence $(\alpha_n) \subset (-1, 1)$, every input v and zero initial state,*

$$\text{OS: } y_n = \sum_{r=0}^n \hat{g}_r(\alpha_n) v_{n-r}, \quad \text{IS: } y_n = \sum_{r=0}^n \hat{g}_r(\alpha_{n-r}) v_{n-r}. \quad (19)$$

The output-scheduled bank is therefore the A-type (2), and the input-scheduled bank the B-type (3), with the realized weights $\hat{g}_r$ in place of g_r .

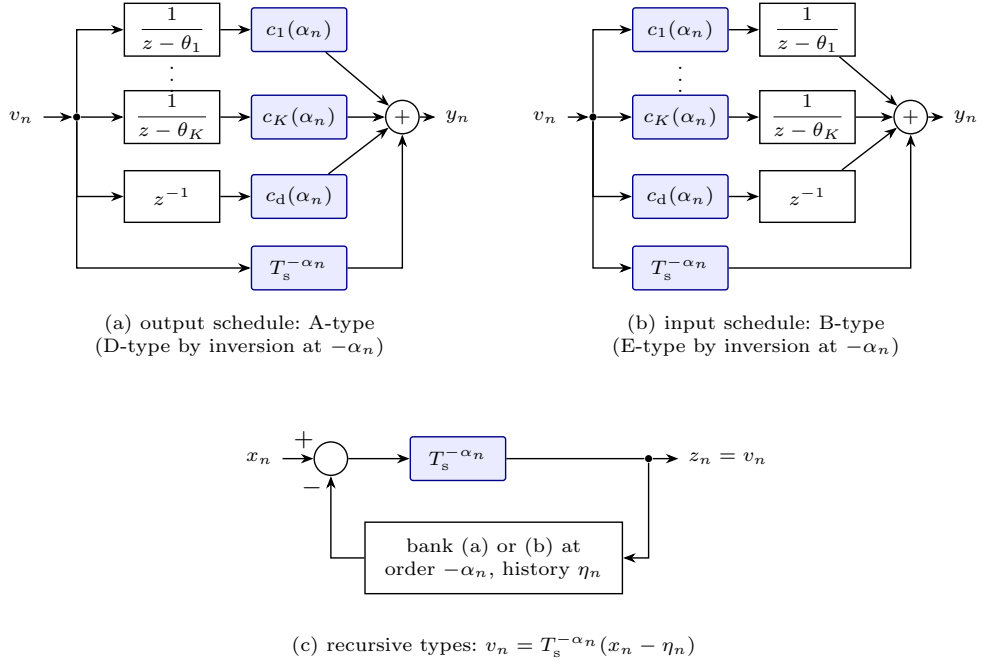


Fig. 1 Fixed-pole bank. The poles θ_k and the delay z^{-1} are common to every order; only the shaded weights depend on α_n . (a) Order in the output weights. (b) Order in the input weights. (c) The recursive types: the history η_n of bank (a) or (b), run at order $-\alpha_n$, is removed and the result is scaled by $1/\hat{g}_0(-\alpha_n) = T_s^{-\alpha_n}$. The feedthrough of the bank at order $-\alpha_n$ is $T_s^{\alpha_n}$, which is why the gain in (c) is $T_s^{-\alpha_n}$

Proof In OS, $s_k[n] = \sum_{r=1}^n \theta_k^{r-1} v_{n-r}$ and $s_d[n] = v_{n-1}$. Hence $\eta_n = \sum_{r \geq 1} (\delta_{r1} c_d(\alpha_n) + \sum_k c_k(\alpha_n) \theta_k^{r-1}) v_{n-r} = \sum_{r \geq 1} \hat{g}_r(\alpha_n) v_{n-r}$. In IS, $s_k[n] = \sum_{r=1}^n \theta_k^{r-1} c_k(\alpha_{n-r}) v_{n-r}$ and $s_d[n] = c_d(\alpha_{n-1}) v_{n-1}$, so $\eta_n = \sum_{r \geq 1} \hat{g}_r(\alpha_{n-r}) v_{n-r}$. $\square$

The state is never modified when the order changes, and no switching term appears. Accuracy is set by the frozen-order design alone:

Corollary 3 Let $\varepsilon_R = \sup_a \max_{1 \leq r < R} |\hat{g}_r(a)/g_r(a) - 1|$ over the design range of orders. For $n < R$,

$$\begin{aligned}
 OS: \quad & \left| y_n - (\mathbf{A} \Delta^\alpha v)_n \right| \leq \varepsilon_R \sum_{r=1}^n |g_r(\alpha_n)| |v_{n-r}|, \\
 IS: \quad & \left| y_n - (\mathbf{B} \Delta^\alpha v)_n \right| \leq \varepsilon_R \sum_{r=1}^n |g_r(\alpha_{n-r})| |v_{n-r}|.
 \end{aligned} \tag{20}$$

The bound is relative to the weights, not to the output. For derivatives the weights nearly cancel ($\sum_r g_r(a) = 0$ for $a > 0$). Relative output errors for smooth inputs can therefore exceed ε_R (Sect. 7.5).

Proposition 4 (Recursive types by inversion) *Run the OS (IS) bank at the order sequence $-\alpha$ and let η_n be its history. The causal recursion*

$$v_n = \frac{x_n - \eta_n}{\hat{g}_0(-\alpha_n)} = T_s^{-\alpha_n}(x_n - \eta_n), \quad z_n = v_n, \quad (21)$$

gives $z = \hat{W}_A(-\alpha)^{-1}x$ (respectively $\hat{W}_B(-\alpha)^{-1}x$), where $\hat{W}$ is (6) with the realized weights. It thus realizes the D-type (E-type) with realized weights, and the duality (7) holds exactly in exact arithmetic:

$$\hat{A}_\Delta^{-\alpha} \hat{D}_\Delta^\alpha = \hat{D}_\Delta^\alpha \hat{A}_\Delta^{-\alpha} = I, \quad \hat{B}_\Delta^{-\alpha} \hat{E}_\Delta^\alpha = \hat{E}_\Delta^\alpha \hat{B}_\Delta^{-\alpha} = I.$$

Each sample costs $O(K)$, the same as the forward types.

Proof By Proposition 2, row n of $\hat{W}_A(-\alpha)v = x$ is $\hat{g}_0(-\alpha_n)v_n + \eta_n = x_n$. Since $\hat{g}_0(-\alpha_n) = T_s^{\alpha_n} \neq 0$, the triangular system is solved forward by (21). The IS case is identical. $\square$

Remark 1 (VO relaxation equations) The same structure solves $T^\top \Delta^\alpha y + \lambda y = u$ for $T \in \{A, B, D, E\}$ at $O(K)$ per sample:

- A and B: $y_n = (u_n - \eta_n)/(\hat{g}_0(\alpha_n) + \lambda)$;
- D and E: write $y = \hat{W}(-\alpha)z$ with $z = u - \lambda y$, which gives $y_n = (\hat{g}_0(-\alpha_n)u_n + \eta_n)/(1 + \lambda \hat{g}_0(-\alpha_n))$, the bank being driven by $z_n = u_n - \lambda y_n$.

A direct triangular solve of the GL equations costs $O(N^2)$.

Corollary 3 bounds each output sample by a weighted sum of past inputs. Because every row of the A-type error matrix and every column of the B-type error matrix holds a single order, the same weight errors also bound the realized operators as a whole, for every order sequence. Let $\|\cdot\|_p$ denote the induced ℓ^p norm of an $N \times N$ matrix and $|M|$ the entrywise absolute value.

Theorem 5 (Operator-norm bounds) *Let $\mathcal{S} \subset (-1, 1)$ be a set of orders, $N \leq R$, and $e_r(a) = \hat{g}_r(a) - g_r(a)$ for $1 \leq r < N$. Put*

$$\rho_{\text{row}} = \sup_{a \in \mathcal{S}} \sum_{r=1}^{N-1} |e_r(a)|, \quad \rho_{\text{col}} = \sum_{r=1}^{N-1} \sup_{a \in \mathcal{S}} |e_r(a)|. \quad (22)$$

For every order sequence with values in $\mathcal{S}$, let $E_T = \hat{W}_T - W_T$ be the error of the realized operator of type T .

- $\|E_A\|_\infty \leq \rho_{\text{row}}$, $\|E_A\|_1 \leq \rho_{\text{col}}$, $\|E_B\|_1 \leq \rho_{\text{row}}$, $\|E_B\|_\infty \leq \rho_{\text{col}}$, and $\|E_T\|_2 \leq (\rho_{\text{row}}\rho_{\text{col}})^{1/2}$ for $T \in \{A, B\}$. If $\mathcal{S}$ is finite, each of the four bounds is attained by some order sequence.
- $\|E_T\|_p \leq \varepsilon_R \|W_T\|_p$ for $1 \leq p \leq \infty$ and $T \in \{A, B\}$. In particular, $\|E_T\|_p \leq \varepsilon_R \|W_T\|_p$ for $p \in \{1, \infty\}$: the relative error of the operator, and hence the worst-case relative output error over all inputs, never exceeds the relative weight error.

(c) For $(T, F) \in \{(D, A), (E, B)\}$, let E_F^- be the error of $\hat{W}_F(-\alpha)$ and $q = \|W_T\|_p \|E_F^-\|_p$. If $q < 1$, then $\|E_T\|_p \leq \frac{q}{1-q} \|W_T\|_p$. The same holds with $\hat{q} = \|\hat{W}_T\|_p \|E_F^-\|_p$ and $\|\hat{W}_T\|_p$ in place of q and $\|W_T\|_p$. If all orders are negative and $\hat{g}_1(-\alpha_n) < 0$ for all n , then $\hat{W}_T \geq 0$ entrywise, and $\|\hat{W}_T\|_\infty$ is the peak of the realized step response.

The proof is given in Appendix C. Part (a) uses the error kernel itself and does not depend on how the order sequence is chosen. Part (c) gives an a-posteriori certificate for the recursive integrals: one step response of the bank ($O(NK)$) and the constants (22) at the opposite orders.

Remark 2 (Conditioning of the recursive types) By (b), $q \leq \varepsilon_R \kappa_p$ with $\kappa_p = \|W_F(-\alpha)\|_p \|W_F(-\alpha)^{-1}\|_p$. For a constant order $a \in (0, 1)$, the partial sums of $(1-z)^{\pm a}$ give

$$\kappa_\infty = (2 - w_{N-1}(a-1)) w_{N-1}(-a-1) \approx \frac{2N^a}{\Gamma(1+a)},$$

which is 71 for $a = 0.5$ and $N = 1000$. For a recursive integral this amplification is real. Its step response is set by the DC gain of the inverted derivative bank, which is small (about $T_s^{-a} N^{-a} / \Gamma(1-a)$) and carries an absolute error of order $\varepsilon_R T_s^{-a}$. A recursive integral with relative operator error ε therefore needs a bank designed for about $\varepsilon / \kappa_\infty$. For recursive derivatives the bound is conservative; the measured errors stay at the level of ε_R (Sect. 7.6).

The fixed-pole structure is not only sufficient for definition consistency. Some version of it is also necessary:

Theorem 6 (Necessity) Let $\Sigma_i = (F_i, b_i, c_i^\top, d_i)$, $i = 1, 2$, be minimal realizations of the same dimension S of two frozen-order operators. The switched system runs Σ_1 for $n < m$ and Σ_2 for $n \geq m$, where $m \geq S+1$, and the state is mapped as $x[m] \leftarrow \Phi x[m]$ at the switch.

- (a) For every input, the output for $n \geq m$ equals the response of Σ_2 to the entire input (*A-type behaviour*) if and only if $F_2 = \Phi F_1 \Phi^{-1}$ and $b_2 = \Phi b_1$.
- (b) For every input, the output for $n \geq m$ equals the response of Σ_1 to the inputs before m plus the response of Σ_2 to the inputs from m on (*B-type behaviour*) if and only if $F_2 = \Phi F_1 \Phi^{-1}$ and $c_2^\top = c_1^\top \Phi^{-1}$.

In both cases F_1 and F_2 are similar, so the two realizations have the same poles.

The proof is given in Appendix A. The same statement holds in continuous time with $e^{F_i t}$ in place of F_i^n .

Corollary 7 If the poles of a frozen-order realization depend on the order, as in the Oustaloup family [12], then no state map realizes the A- or B-type exactly at a switch between orders with different poles. In particular the hot swap $\Phi = I$ does not. The fixed-pole bank satisfies (a) with $\Phi = I$ in OS, where (F, b) does not depend on the order, and (b) with $\Phi = I$ in IS, where $(F, c^\top)$ does not depend on the order.

Remark 3 (Approximate maps) Theorem 6 rules out exact maps, not approximate ones. A least-squares Φ fitted to an input ensemble can make the definitional error small in floating point (Sect. 7.3), because the reachable states of both realizations concentrate in a low-dimensional subspace. Such a map, however:

- is dense, costing S^2 multiply-accumulates (MACs) at every switch;
- is specific to each pair of orders, needing $2(P-1)$ maps for the $P-1$ neighbouring pairs of a grid of P order levels;
- depends on the statistics of the input.

State-update strategies for time-varying recursive filters suppress the transients of a coefficient change [46, 47]. By Theorem 6, no such update can make a moving-pole realization definition-consistent.

5 Error model, design rule and inverse stability

The design rule chooses h , u_{lo} , u_{hi} and hence K from a tolerance ε , a memory length R (in samples) and an order range $[\alpha_{\min}, \alpha_{\max}]$. In discrete time the metric is the relative weight error ε_R of Corollary 3. In continuous time it is $\sup_{\omega_1 \leq \omega \leq \omega_2} |H_\alpha(j\omega)/(j\omega)^\alpha - 1|$. Every term below follows from the representation, and no constant is fitted.

5.1 Discrete time

With the tail closure, the realized weights are the infinite trapezoidal rule applied to (14), apart from two tail residuals. The error therefore has three terms.

Quadrature.

The integrand of (14) is analytic in a strip around the real axis, so the error of the infinite trapezoidal rule is the aliasing of its Fourier transform at $2\pi/h$ (Poisson summation) [51]. For large r the integrand is $e^{(1+a)x}e^{-(r-a)e^x}$ to leading order. The modulus of its transform at $2\pi/h$ is $|\Gamma(1+a-2\pi i/h)|$ relative to $\Gamma(1+a)$, and Stirling's formula gives (Supplementary Sect. S7)

$$E_q^\infty(a, h) = \frac{2\sqrt{2\pi} (2\pi/h)^{a+1/2} e^{-\pi^2/h}}{\Gamma(1+a)}. \quad (23)$$

For small r the alias amplitude is evaluated directly from the representation. The relative errors of the infinite trapezoidal sum at the grid offsets 0 and $h/4$ are the cosine and sine components of the first alias, and their root sum of squares is its amplitude. We use $E_q = \max(E_q^\infty, \max_{r \in \{1, 2, 3, 5, 8\}} E_q^{(r)})$.

Lower tail.

The virtual nodes below u_{lo} are matched to θ_0 at $r = 1$. At lag r their mismatch is $\sum_j h\varphi_a(u_{-j})(\theta_{-j}^{r-1} - \theta_0^{r-1}) \approx (r-1)u_{\text{lo}}^{2+a}H_h(1+a, 2+a)$ times $-S_a T_s^{-a}$. Relative to

Table 2 Continuous-time error terms (relative frequency-response error on $\omega_1 \leq \omega \leq \omega_2$), with $S = \sin(\pi a)/\pi$ and G_h, H_h from (12)

Term	integral, $\alpha = -a < 0$	derivative, $\alpha = a > 0$
quadrature (first alias)	$2 \sin(\pi a) e^{-\pi^2/h}$	$2 \sin(\pi a) e^{-\pi^2/h}$
lower tail	$S(\xi_{\text{lo}}/\omega_1)^{2-a} H_h(1-a, 2-a)$	$S(\xi_{\text{lo}}/\omega_1)^{1+a} G_h(1+a)$
upper tail	$S(\omega_2/\xi_{\text{hi}})^{1+a} G_h(1+a)$	$S(\omega_2/\xi_{\text{hi}})^{2-a} H_h(1-a, 2-a)$

$|w_r(a)| = |S_a|B(r-a, 1+a)$ this gives the residual

$$E_{\text{lo}} = \frac{(r-1) u_{\text{lo}}^{2+a} H_h(1+a, 2+a)}{B(r-a, 1+a)}, \quad \text{worst at } r = R-1. \quad (24)$$

It grows with the lag, so it is the term that ties u_{lo} to the memory length.

Upper tail.

The delay mode matches the nodes above u_{hi} at $r = 1$, so the first residual appears at $r = 2$:

$$E_{\text{hi}} = \frac{1}{B(2-a, 1+a)} \sum_{j \geq 1} h u_j e^{-(2-a)u_j} (1 - e^{-u_j})^a, \quad u_j = u_{\text{hi}} e^{jh}. \quad (25)$$

5.2 Continuous time

The same reasoning applied to (10) gives the terms of Table 2. The quadrature term is $2|\sin \pi a|e^{-\pi^2/h}$. The lower and upper tails exchange their forms between integrals and derivatives, as a consequence of the substitution $\xi \rightarrow 1/\xi$ in (10).

5.3 Rule

The tolerance is split as $\varepsilon/2$ for the quadrature term and $\varepsilon/4$ for each tail. Over a grid of orders covering the design range, the rule proceeds in four steps:

1. Choose the largest h with $\max_a E_q(a, h) \leq \varepsilon/2$. In continuous time this is the closed form $h = \pi^2 / \ln(4 \max_a |\sin \pi a|/\varepsilon)$; in discrete time it is a one-dimensional root.
2. For each order, solve the lower-tail condition in closed form, since it is a power law in u_{lo} (in ξ_{lo}/ω_1 in continuous time), and take the smallest value.
3. For each order, find u_{hi} (or ξ_{hi}) by a monotone root and take the largest value.
4. Set $K = \lceil \ln(u_{\text{hi}}/u_{\text{lo}})/h \rceil + 1$; the realized spacing then does not exceed the h of step 1.

5.4 Inverse stability and the DC floor

The recursive types (21) run the frozen-order inverse $1/\hat{H}$. Its dynamics matrix is $F - bc^\top/\hat{g}_0$ with $F = \text{diag}(\theta_0, \dots, \theta_{K-1}, 0)$, $b = \mathbf{1}$ and $c = (c_0, \dots, c_{K-1}, c_d)$. For the

input schedule the dynamics matrix is $F - cb^\top/\hat{g}_0$, the transpose structure. Both have the characteristic polynomial $N(z)/\hat{g}_0$ with

$$N(z) = z \prod_k (z - \theta_k) \hat{H}(z).$$

The inverse is therefore stable if and only if all roots of N lie in the open unit disc.

Proposition 8 (Inverse stability) *Let $\hat{H}(z) = g_0 + c_d/z + \sum_k c_k/(z - \theta_k)$ with $g_0 > 0$ and distinct $\theta_k \in (0, 1)$.*

- (i) *If $c_k > 0$ for all k and $c_d \geq 0$ (integral bank), all roots of N are real and $1/\hat{H}$ is stable if and only if $\hat{H}(-1) > 0$.*
- (ii) *If $c_k < 0$ for all k (derivative bank), all roots of N are real and $1/\hat{H}$ is stable if and only if $\hat{H}(1) > 0$ and either $c_d \leq 0$ or $\hat{H}(-1) > 0$.*
- (iii) *If $|\hat{g}_r/g_r - 1| \leq 1/3$ for $r = 1, 2$, then $\hat{H}(-1) > 0$.*

Hence, for any bank whose first two weights are accurate to 1/3 (which every design of the rule satisfies by a wide margin), an integral bank always has a stable inverse, and a derivative bank has one if and only if its realized DC gain $\hat{H}(1) = \sum_r \hat{g}_r$ is positive.

The proof (Appendix B) has two ingredients:

- interlacing: between consecutive poles with residues of equal sign there is exactly one root;
- monotonicity of $z\hat{H}(z)$ on $z > \max_k \theta_k$ for a derivative bank, which locates the root closest to $z = 1$.

The exact value of $\hat{H}(-1)$ is $\sum_r g_r(-1)^r = 2^a T_s^{-a}$. On the 25 orders checked in each of the 200 discrete-time holdout designs of Sect. 7.4, with and without the floor, the realized value is within 0.06 % of it.

For an exact GL derivative, $\sum_r g_r(a) = 0$, so the inverse has a pole on the unit circle in the limit. The finite memory of the bank adds a small positive DC term, approximately

$$S_a T_s^{-a} u_{lo}^a / (a(1+a)),$$

the part of the density below u_{lo} not represented by the lumped closure. For large orders and small u_{lo} this term is smaller than the quadrature error of the DC sum, and $\hat{H}(1)$ can become negative. We therefore apply a DC floor: if $\hat{H}(1) < H_{\min} = 0.01 |S_a| T_s^{-a} u_{lo}^a / (a(1+a))$, the delay residue is raised by $H_{\min} - \hat{H}(1)$. This changes only $\hat{g}_1$, by about the DC quadrature error, and makes every inverse stable by Proposition 8.

Remark 4 (Signs instead of eigenvalues) The dominant pole of a derivative-bank inverse lies extremely close to $z = 1$, and general eigenvalue solvers are unreliable there. In one holdout design (order 0.95, $K = 35$), a 50-digit bisection places the dominant pole at $1 - 2.9 \times 10^{-12}$, while a double-precision eigenvalue routine reports modulus $1 + 4.5 \times 10^{-9}$. The signs of $\hat{H}(1)$

and $\hat{H}(-1)$, in contrast, are computed to full relative precision, and in fixed point they are evaluated exactly (Sect. 9).

6 Beyond Grünwald–Letnikov: completely monotone kernel families

Sections 3–5 used two properties of the GL weights. For each frozen order they are a superposition of geometric sequences with a density of one sign, and this density is smooth on a logarithmic scale. Both properties hold for a much larger class of kernels.

Definition 2 (CM family) A family of causal kernels $\{k_a\}_{a \in \mathcal{A}}$ is a *completely monotone (CM) family* if $k_a(0) \neq 0$ and, for $r \geq 1$,

$$k_a(r) = \sigma_a \int_0^\infty e^{-(r-1)u} m_a(u) du, \quad \sigma_a \in \mathbb{R}, \quad m_a \geq 0. \quad (26)$$

Its A- and B-type operators are (2) and (3) with $k_a(r)$ in place of $g_r(a)$, and its D- and E-types are defined by the duality (7).

With $\theta = e^{-u}$, (26) states that $(k_a(r)/\sigma_a)_{r \geq 1}$ is the moment sequence of a nonnegative measure on $(0, 1)$. By Hausdorff’s theorem these are exactly the completely monotone sequences [52]; here the measure has a density, and an atom at $\theta = 0$ is the delay mode. For the GL weights, Lemma 1 gives $\sigma_a = -S_a T_s^{-a}$ and $m_a(u) = e^{-(1-a)u}(1 - e^{-u})^a$. The bank of a CM family is built as in Sect. 3: poles $\theta_k = e^{-u_k}$ with $u_k = u_{10} e^{kh}$, residues $c_k(a) = \sigma_a h u_k m_a(u_k)$, the nodes below the grid lumped onto θ_0 and matched at $r = 1$, and the nodes above it collected in the delay mode.

Theorem 9 (Fixed-pole banks for CM families) *Let $\{k_a\}$ be a CM family and $\hat{k}_a$ the weights realized by its bank.*

- (a) *Propositions 2 and 4, Corollary 3 and Theorems 5 and 6 hold with $k_a(r)$ in place of $g_r(a)$. The parameter a need not be an order. All residues have the sign of σ_a , so parts (i) and (ii) of Proposition 8 apply to every member.*
- (b) *Assume that there are constants $0 < d < \pi/2$, $0 < \beta_- \leq \beta_+$, $\gamma > 0$, $0 < c_- \leq c_+$, C_∞ and $u_* > 0$ such that, for all $a \in \mathcal{A}$:*
 - (H1) *$f_{a,r}(x) = e^x m_a(e^x) e^{-(r-1)e^x}$ is analytic in the strip $|\operatorname{Im} x| < d$ and tends to zero as $|\operatorname{Re} x| \rightarrow \infty$ in it, and*

$$L_d = \sup_{a \in \mathcal{A}, r \geq 1, |y| < d} \frac{\int_{\mathbb{R}} |f_{a,r}(x + iy)| dx}{\int_{\mathbb{R}} f_{a,r}(x) dx} < \infty;$$

- (H2) *$c_- u^{\beta_a - 1} \leq m_a(u) \leq c_+ u^{\beta_a - 1}$ on $(0, u_*]$, with $\beta_a \in [\beta_-, \beta_+]$;*
- (H3) *$m_a(u) \leq C_\infty e^{-\gamma u}$ on $[u_*, \infty)$.*

Let $\vartheta = \min_{\beta_- \leq \beta \leq \beta_+} \int_0^{u_*} t^{\beta-1} e^{-t} dt$, $C_{\text{lo}} = c_+/(c_- \beta_- \vartheta)$ and $C_{\text{hi}} = C_\infty/((1 + \gamma)c_- \vartheta)$. Then, for $u_{\text{lo}} \leq u_*$, $u_{\text{hi}} \geq \max(u_*, 1, \beta_+ \ln 2)$ and $1 \leq r < R$,

$$\left| \frac{\hat{k}_a(r)}{k_a(r)} - 1 \right| \leq \frac{2L_d}{e^{2\pi d/h} - 1} + C_{\text{lo}}((R-1)u_{\text{lo}})^{1+\beta_a} + C_{\text{hi}} e^{-(1+\gamma)u_{\text{hi}}}. \quad (27)$$

(c) For $0 < \varepsilon \leq \min(1, 4C_{\text{lo}})$ let $h = 2\pi d/\ln(1 + 4L_d/\varepsilon)$, $u_{\text{lo}} = \min(u_*, (\varepsilon/4C_{\text{lo}})^{1/(1+\beta_-)}(R-1))$, $u_{\text{hi}} = \max(u_*, 1, \beta_+ \ln 2, \ln(4C_{\text{hi}}/\varepsilon)/(1+\gamma))$ and $K = \lceil \ln(u_{\text{hi}}/u_{\text{lo}})/h \rceil + 1$. Then the relative weight error is at most ε for every $a \in \mathcal{A}$ and $1 \leq r < R$, and

$$K = O(\ln(1/\varepsilon) \ln(R/\varepsilon)) \quad (\varepsilon \rightarrow 0, R \rightarrow \infty), \quad (28)$$

uniformly in a .

The proof is given in Appendix D. Part (a) holds because the earlier proofs use only the state equations of the bank and the form $\hat{k}_a(r) = \delta_{r1}c_d(a) + \sum_k c_k(a)\theta_k^{r-1}$. Part (b) combines the error bound of the trapezoidal rule for functions analytic in a strip [51, Thm. 5.1] with the two tail residuals of Sect. 5. For the GL weights all constants are explicit.

Lemma 10 (GL constants) *Let $|a| \leq \bar{a} < 1$ on $\mathcal{A}$. The GL family satisfies (H1)–(H3) for every $0 < d < \pi/2$, with $\beta_a = 1 + a$, $\gamma = 1 - \bar{a}$, $u_* = 1$, $c_- = (1 - e^{-1})e^{-2}$, $c_+ = C_\infty = (1 - e^{-1})^{-1}$ and*

$$L_d \leq \left(\frac{2}{1 - e^{-2}} \right)^{\bar{a}} (\cos d)^{-(1+\bar{a})}. \quad (29)$$

The proof is in Appendix D. With these constants, part (c) of Theorem 9 gives a bank for the GL family without any error model specific to it; Sect. 7.7 compares it with the rule. The quadrature term (23) of the rule is the sharper limit $d \rightarrow \pi/2$ for this family: it has π^2/h in the exponent and an algebraic prefactor. With $h \approx \pi^2/\ln(1/\varepsilon)$, (28) becomes $K \approx \ln(1/\varepsilon) \ln(R/\varepsilon)/\pi^2$ to leading order. A fit over 35 rule designs gives 0.106 for this coefficient, against $1/\pi^2 = 0.101$ (Sect. 7.7).

Example 1 (Distributed and time-varying distributed order) For a probability measure ν on orders of one sign, $k(r) = \int g_r(a) d\nu(a)$ is again a CM kernel, with density $\int \sigma_a m_a d\nu(a)$. Its bank has the same poles and the residues $\int c_k(a) d\nu(a)$. Because all weights have one sign, the relative weight error is at most $\sup_a \varepsilon_R(a)$. If the distribution ν_n changes with time, the output schedule weights the whole memory with ν_n (A-type), and the input schedule weights each sample with the distribution at its entry (B-type). Both still cost $O(K)$ per sample. For distributions of mixed sign the structural statements remain exact, and the error is relative to $\int |g_r(a)| d\nu(a)$.

Example 2 (Tempering) For a fixed $\lambda > 0$, the tempered Grünwald weights $e^{-\lambda r} g_r(a)$ [53] (up to a change of the lag-zero weight) have the density $e^{-\lambda} m_a(u - \lambda)$ on $u > \lambda$. The bank

with poles $e^{-\lambda}\theta_k$, residues $e^{-\lambda}c_k(a)$ and delay residue $e^{-\lambda}c_d(a)$ realizes $e^{-\lambda r}\hat{g}_r(a)$ exactly. Its relative weight error, its type and its design rule are therefore those of the untempered bank. A tempering parameter that varies in time moves the poles. By Theorem 6, switching it with a fixed state then cannot give the A- or B-type exactly. A fixed bank for a range of λ is possible, but its density has a branch point at $u = \lambda$, (H1) fails, and the trapezoidal rule converges only algebraically.

Remark 5 (Continuous time) By Bernstein’s theorem, the completely monotone functions are the Laplace transforms of nonnegative measures [52]. For kernels $k_a(t) = \sigma_a \int_0^\infty e^{-\xi t} m_a(\xi) d\xi$ the continuous-time bank of Sect. 3 gives the same structural statements, with e^{Ft} in place of F^n . Examples are the power laws of (10), their distributed-order mixtures, and the Prabhakar kernels in their completely monotone parameter range [54].

7 Numerical results

7.1 Setup

Unless stated otherwise, $T_s = 1$ ms. Discrete-time references are the literal definitions (2)–(5), evaluated in $O(N^2)$ and checked against (7). Continuous-time references are (8) and (9). Errors are relative RMS errors over the post-switch window, or over the whole record.

To separate what switching does from how well each frozen-order member approximates s^α , we also report a *definitional error*: the distance of a switched realization from its *own* A-, B-, D- or E-type, built from the same family of frozen-order systems. For a piecewise-constant order with value set V and masks $M_v = [\alpha = v]$, the own A-type is $y_n = (R_{\alpha_n} u)_n$ and the own B-type is $y = \sum_{v \in V} R_v(u \cdot M_v)$. The own D- and E-types follow from (7).

The baselines are Oustaloup approximations [12] on $[10^{-3}, 10^5]$ rad/s with $N = 3, \dots, 20$ (state count $S = 2N + 1$). They are used in modal (parallel) and first-order-cascade form, discretized by zero-order hold, with the state kept on every coefficient swap. Seeds are fixed, and every table and figure is produced by one script per experiment.

7.2 Which definition does a switched realization track?

Continuous time.

Figure 2 and Supplementary Table S1 show a unit step with one order switch at $t = 2$ s, at equal state count. The fixed-pole bank with output schedule tracks the A-type and the one with input schedule tracks the B-type, with errors down to 10^{-4} – 10^{-5} . Before the switch, all realizations have similar errors (10^{-3} – 10^{-4}), so the differences come from the switch. The hot-swapped parallel Oustaloup approaches the A-type only slowly. The cascade form tracks neither type: its error stays between 0.16 and 4.6 and does not decrease with S .

One negative result motivates the discrete-time bank. The continuous-time bank with input schedule fails for a smooth *derivative* order (relative error about 12), because modes faster than $1/T_s$ respond to every step of the sampled order.

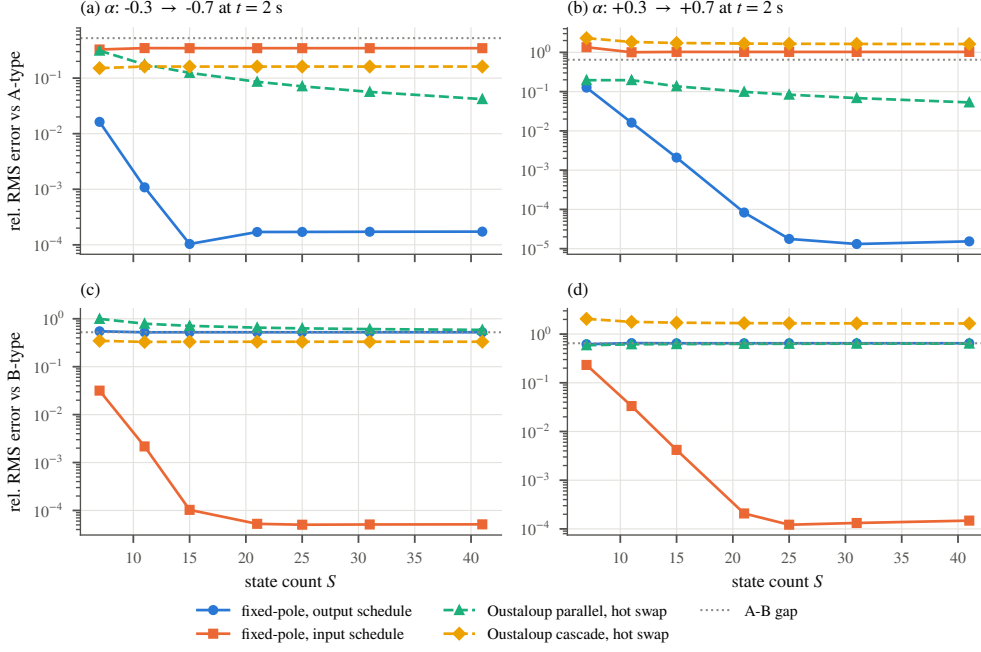


Fig. 2 Continuous time, unit step, one order switch at $t = 2$ s. Relative RMS error after the switch against the A-type (top) and B-type (bottom) closed forms versus state count. The dotted line is the distance between the two definitions. Every series has its own marker and line style

Table 3 Discrete time, piecewise-constant orders: definitional error (distance from the realization's own type). DFP is the discrete fixed-pole bank; the Oustaloup forms have $N = 15$ ($S = 31$)

Realization	vs own A-type	vs own B-type
DFP, output schedule	0 (exact)	equals the A-B gap
DFP, input schedule	equals the A-B gap	$\leq 10^{-14}$
Oustaloup parallel, hot swap	4×10^{-6} to 0.13	7×10^{-3} to 1.0
Oustaloup cascade, hot swap	7×10^{-3} to 3.0×10^3	0.03 to 4.8×10^2

Discrete time.

The discrete bank (32 modes and the delay mode) was compared with the GL definitions over 7 order profiles and 3 inputs (step, white noise, multisine). The total error of the output schedule against the A-type was 2.5×10^{-6} to 2.0×10^{-3} , and that of the input schedule against the B-type 1.9×10^{-6} to 1.6×10^{-3} . Smooth derivative orders are included, so the continuous-time failure disappears. Table 3 gives the definitional errors. For the bank they are zero, or zero to rounding, as Proposition 2 predicts. For the hot-swapped Oustaloup forms they are not, and the cascade form reaches 3×10^3 .

7.3 Can a state map repair a moving-pole realization?

We fitted least-squares maps Φ for six single switches on mixed input ensembles and tested them on fresh inputs and on a unit step (details in Supplementary Sect. S2). In float64 the worst A-type definitional error fell from 0.14 (parallel form, hot swap) and 1.1×10^4 (cascade form) to below 10^{-6} at $S = 31$. The map, however, is a dense $S \times S$ matrix with norm 10^3 to 5×10^4 , specific to each pair of orders. Rounded to 24 bits, it loses 3.5 to 4.3 orders of magnitude of accuracy in the parallel form and 7.4 to 9.1 in the cascade form ($S \geq 21$). For the 1024 order levels of Sect. 9, maps between neighbouring levels alone would need about 56 Mbit, against 0.78–1.08 Mbit for the whole coefficient table of the bank, which needs no map.

7.4 Design rule on a sealed holdout

Term validation.

Each term of Sect. 5 was first isolated by making the other two negligible (Fig. 3a,b). The ratio of measured to predicted error was 0.49–1.05 in discrete time ($n = 45$) and 0.99–1.01 in continuous time ($n = 66$). The model never underestimates a term by more than 5%.

Holdout.

We drew 400 random specifications (200 discrete, 200 continuous) from a fixed seed:

- $\varepsilon \in [10^{-5}, 10^{-2}]$;
- $R \in [10^2, 10^5]$, or bands of 0.5 to 6 decades;
- random order ranges inside $[-0.95, 0.95]$.

The specifications were written with their SHA-256 digest before any evaluation and then evaluated once. Every specification was met: 200 of 200 in both domains. The one-sided 95% Clopper–Pearson lower bound [55] on the pass rate is 98.5% for each domain. The achieved error was a median 0.41 (discrete) and 0.50 (continuous) of the requested ε , with maxima 0.67 and 0.72. Let $K_{\min}$ be the smallest K on the same grid endpoints that still meets ε , found by bisection. The rule exceeded it by a median of one mode and at most three. Table 4 lists typical sizes: each decade of memory adds about three modes, and each decade of accuracy seven to ten.

7.5 Four types from one bank

Definitions.

The literal recursions (4), (5) agree with $W_A(-\alpha)^{-1}$ and $W_B(-\alpha)^{-1}$ to 1.4×10^{-13} . The dual compositions ($A \circ D$, $D \circ A$, $B \circ E$, $E \circ B$) return the identity to 10^{-12} . The non-dual compositions ($A \circ A$, $B \circ B$, $A \circ B$, $B \circ A$, $A \circ E$, $B \circ D$, $D \circ E$, $D \circ D$) differ from the identity by 10^{-2} to 2.4×10^3 .

Bank.

A bank designed for $\varepsilon = 10^{-5}$, $|\alpha| \leq 0.95$ and $R = 4000$ ($K = 41$) was run in all four types over 5 order profiles and 3 inputs (Table 5). The error against the literal

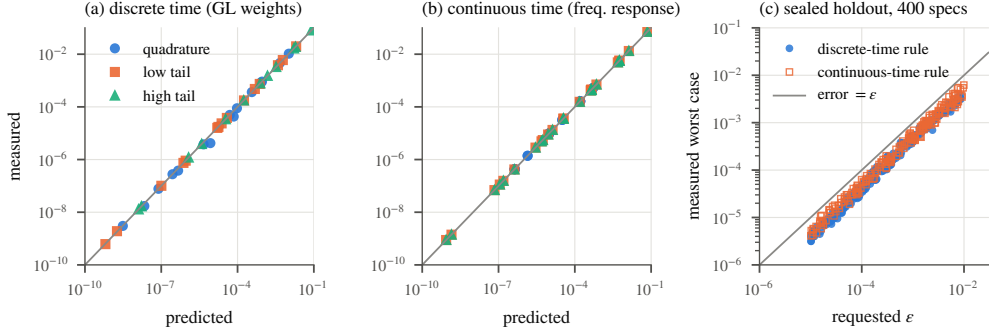


Fig. 3 Analytic error model. (a, b) Each term isolated, measured against predicted. (c) Sealed holdout: measured worst-case error against the requested tolerance for 400 random specifications

Table 4 Bank size K from the rule for $\alpha \in [-0.9, 0.9]$. Discrete time: memory length R in samples (plus the delay mode). Continuous time: band width in decades

ϵ	discrete time, R			continuous time, band		
	10^3	10^4	10^5	2 dec.	4 dec.	6 dec.
10^{-2}	14	17	19	11	14	16
10^{-3}	21	24	27	18	22	26
10^{-4}	29	32	36	27	32	37
10^{-5}	38	42	46	38	44	50

Table 5 One bank ($K = 41$) in four types: error against the literal GL definitions (15 cases), and against the bank's own type (the 12 piecewise-constant cases)

Type	vs literal definition	vs own type (definitional)
A	9.5×10^{-8} to 6.9×10^{-5}	$\leq 2.6 \times 10^{-14}$
B	3.1×10^{-8} to 1.2×10^{-5}	$\leq 1.0 \times 10^{-14}$
D	2.0×10^{-8} to 6.5×10^{-5}	$\leq 2.3 \times 10^{-14}$
E	2.2×10^{-8} to 5.5×10^{-5}	$\leq 2.8 \times 10^{-13}$

definitions is at most 6.9×10^{-5} . The definitional error is at rounding level, and the duality holds in the bank to 2×10^{-14} . The largest output errors occur for derivatives with smooth inputs, where the weights cancel (Corollary 3).

Inverse stability.

Over the 200 discrete-time holdout designs, checked at 25 orders spanning each design range, the inverse was stable at every order in 200/200 designs with the DC floor and

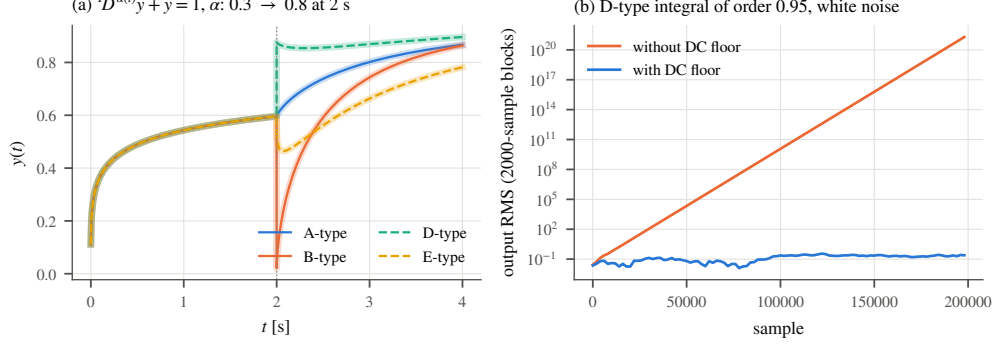


Fig. 4 (a) Relaxation equation of all four types with one bank; thin lines are the bank, wide light lines the triangular GL solutions. (b) D-type integral of order 0.95 (inverse of the 0.95 derivative bank) on white noise, with and without the DC floor

in 101/200 without it. Figure 4b shows a D-type integral of order 0.95 driven by white noise for 2×10^5 samples ($\varepsilon = 10^{-2}$, $R = 1000$, $K = 15$). Without the floor the output RMS grows to 1.9×10^{21} ; with the floor it stays below 0.36.

Which type does a hot swap implement?

Measured against their own A-, B-, D- and E-types over eight cases, the fixed-pole bank is always closest to A (output schedule) or B (input schedule), at distances below 2.1×10^{-14} . The parallel Oustaloup form is closest to the A-type, at distances up to 0.42, and the cascade form has no consistent type (Supplementary Sect. S3).

Relaxation equations.

For ${}^T \Delta^\alpha y + \lambda y = u$ (Remark 1; $K = 40$, $n = 4000$), the bank matched the triangular GL solutions to 1.7×10^{-5} ($\lambda = 1$) and 4.9×10^{-7} ($\lambda = 100$). The solutions of different types differed by up to 0.40 for $\lambda = 1$ (Fig. 4a), so the choice of type changes the solution visibly. For $\lambda = 100$ the term λy dominates and the types differ by at most 0.02.

7.6 Operator-norm error

Two rule designs with $R = N = 1000$ and $|\alpha| \leq 0.95$ were used: $\varepsilon = 10^{-3}$ ($K = 22$) and $\varepsilon = 10^{-6}$ ($K = 49$). For each, the realized and literal operators of all four types were formed as dense 1000×1000 matrices for eleven order sequences:

- the constants ± 0.5 and the constant order with the largest error row sum;
- three single switches, a random piecewise-constant sequence, an independent uniform order at every sample, and a smooth zero crossing;
- two adversarial sequences that take, at every lag, the order with the largest error on a grid of 380 orders.

The matrices reproduce the streaming bank to rounding. Table 6 compares the measured norms with Theorem 5.

Table 6 Theorem 5 on dense operators ($N = 1000$, eleven order sequences). Ratios of measured to bounded quantities; ε_R is taken over the orders of each sequence, except in the last row, where it is the design value

Types	Quantity	$\varepsilon = 10^{-3}$, $K = 22$	$\varepsilon = 10^{-6}$, $K = 49$
A, B	$\ E\ _p/\text{bound (a)}, p = 1, \infty$	0.28–1.00	0.45–1.00
A, B	$\ E\ _2/\text{bound (a)}$	0.13–0.87	0.16–0.89
A, B	$(\ E\ _p/\ W\ _p)/\varepsilon_R, p = 1, \infty$	0.04–0.39	0.03–0.45
D, E, integrals	$\ E\ _\infty/\text{bound (c)}$	0.23–0.31	0.006–0.23
D, E, all	$\max(\ E\ _\infty/\ W\ _\infty)/\varepsilon_R$	22	17

For the A- and B-types no norm exceeded its bound, and the bounds of part (a) were attained to four digits by the constant and the adversarial sequences. The relative operator error in ℓ^1 and ℓ^∞ was at most $0.45 \varepsilon_R$. The error kernel is much smaller than ε_R times the weights: at $\varepsilon = 10^{-3}$, $\rho_{\text{row}} = 0.031$, against $\varepsilon_R \sup_a \sum_r |g_r(a)| = 0.27$. The weight error is relatively largest where the weights are small.

For the recursive integrals (the two sequences with only negative orders), the measured ℓ^∞ error was 0.23–0.31 of the bound of part (c) at $\varepsilon = 10^{-3}$, and the certificate computed from one step response of the bank gave the same values. The amplification predicted by Remark 2 is real: the relative error of the D- and E-types reached 22 times the relative weight error of the design. For recursive derivatives it stayed below ε_R , and the bound was conservative by factors of 14 to 7×10^4 , which is the condition number κ_∞ (71 to 1445 here) times the margin of part (b).

7.7 Completely monotone families

Strip constant and quadrature bound.

For the GL family, L_d was evaluated over 13 orders in $[-0.95, 0.95]$ and lags $r = 1$ to 10^5 . It is 2.0, 7.2, 31.7 and 175 for $d = 0.8, 1.2, 1.4$ and 1.5 , which follows $(\cos d)^{-1.95}$, and stays below the closed-form bound (29) (4.5, 16.1, 70 and 388) by a factor of 2.2. Minimized over d , the quadrature term of (27) exceeded the measured worst error of the infinite trapezoidal rule (13 orders, 10 lags, 8 grid offsets) for $0.3 \leq h \leq 1$: by factors of 1.9 to 3.7 with the measured L_d , and of 4.3 to 7.6 with (29). The sharper GL term (23) agreed with the measurement to within 1%. The general theorem thus loses less than a factor of four against the analysis specific to GL, which costs two to three modes at $K \approx 40$.

Bank size.

Over 35 rule designs with $R = 10^2, \dots, 10^6$ and $\varepsilon = 10^{-2}, \dots, 10^{-8}$ (K from 12 to 90), the least-squares fit

$$K \approx 0.57 + 1.06 \ln(1/\varepsilon) + 0.53 \ln R + 0.106 \ln(1/\varepsilon) \ln(R/\varepsilon)$$

reproduces every K to within 0.7 modes. The coefficient of the product term is within 5% of the value $1/\pi^2$ predicted by Theorem 9.

A-priori design.

Part (c) of Theorem 9 with the constants of Lemma 10, and d chosen to minimize K , gives $K = 36, 58, 67$ and 90 for $(\varepsilon, R) = (10^{-3}, 10^3), (10^{-5}, 10^3), (10^{-5}, 10^5)$ and $(10^{-7}, 10^4)$. The rule gives $22, 39, 47$ and 66 for the same specifications. The measured errors of the theorem-designed banks (without the DC floor) were 13 to 17 times below ε . The general theorem thus yields a guaranteed design that needs no error model specific to GL, at 1.4 to 1.6 times the size of the rule.

Distributed and tempered kernels.

For three time-varying distributed orders (Example 1; uniform densities of width 0.3 whose centre moves smoothly in $[0.15, 0.75]$ or in $[-0.75, -0.15]$, or switches from -0.5 to 0.5), one bank designed for $\varepsilon = 10^{-5}$ ($K = 39$) was run with the averaged residues. Its kernel error was below the largest error of its members in all three cases (at most 2.6×10^{-6}). The relative operator error of the A- and B-types was at most 1.8×10^{-7} , and the streaming bank matched the dense operators to 3×10^{-16} . For the tempered weights with $\lambda = 0.01$ (Example 2), the relative weight error equals that of the untempered bank, 3.877×10^{-6} , to rounding.

7.8 Cost against accuracy

Four ways of realizing the A- and B-types were compared at $T_s = 1$ ms with memory $R = N = 1000$ (Fig. 5). The orders take the $P = 39$ levels $-0.95, -0.90, \dots, 0.95$, which is how a ROM-addressed core sees them (Sect. 9).

- *Fixed poles, analytic rule* (this paper): $\varepsilon = 10^{-2}, \dots, 10^{-7}$, $K = 15$ to 60 .
- *Fixed poles, fitted residues*: a geometric pole grid whose end points are chosen by a grid search, with the residues of each level fitted by a linear program (LP) that minimizes the worst relative weight error over the lags; $K = 6$ to 28 . The LP is solved once without constraints, and once with the residue signs and the DC conditions under which Proposition 8 certifies the inverse.
- *Moving poles*: each level has its own rule-designed grid on a common state count, and the state is kept when the level changes (hot swap).
- *Truncated GL convolution* with exact weights up to lag $L - 1$, $L = 2$ to N .

Accuracy is the relative operator error in ℓ^∞ , which is the worst-case relative output error over all bounded inputs. It was taken over the 39 constant orders, and over eight order sequences: three single switches, a sign change, a fast alternation, random piecewise-constant levels, independent levels, and a smooth crossing. Cost is counted as multiplications per sample ($2K + 2$ for the banks, L for the convolution) and as memory words. The memory is the state plus a coefficient ROM of $P(K + 2) + K$ words for fixed poles, $P(2K + 2)$ for moving poles, and PL for the convolution, which also stores L samples.

Moving poles.

At a constant order, per-level grids reach the accuracy of the common grid with about two thirds of the states ($K = 25$ against 39 at about 10^{-6}). Under switching, their

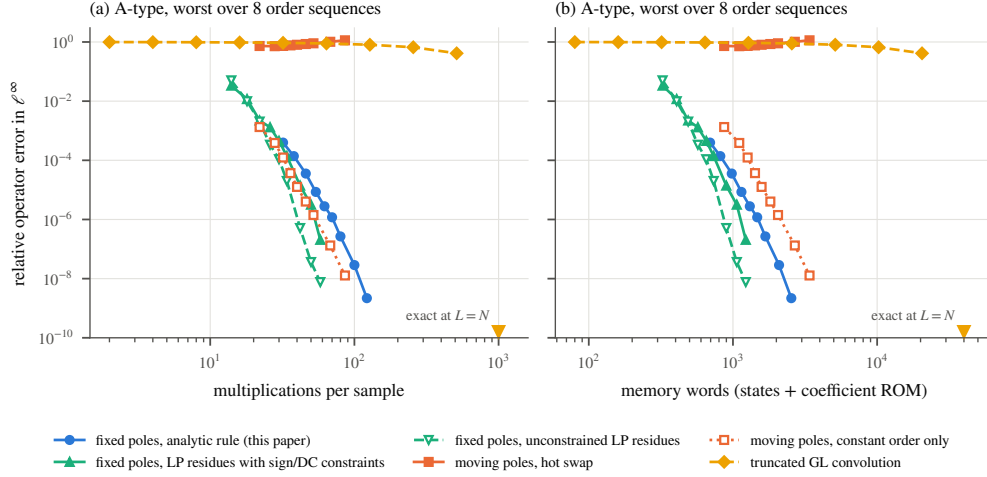


Fig. 5 Accuracy against cost for the A-type ($N = 1000$, 39 order levels). Solid lines: worst relative operator error over eight variable-order sequences; dotted: moving poles at constant order only. The B-type gives the same picture (moving poles: 0.17 to 0.23 under switching)

error is 0.72 to 1.1 for the A-type and 0.17 to 0.23 for the B-type, at every size from $K = 10$ to 42. It does not decrease with K , as Theorem 6 predicts. The Oustaloup forms of Sect. 7.3 behave in the same way.

Truncated convolution.

The integral orders need the whole memory. The worst error is still 0.41 at $L = 512$, and the convolution is exact only at $L = N$, with 1000 multiplications per sample and 39 000 ROM words. The rule-designed bank with $K = 39$ reaches 2.7×10^{-7} with 80 multiplications and about 1700 words.

Fitted residues.

The structure, and with it the type, is the same for every residue design, and the variable-order error of every fixed-pole design stayed below its relative weight error, as Theorem 5(b) states. Unconstrained LP residues need 40 to 50 % fewer poles than the rule ($K = 20$ against 34 for about 10^{-6}). However, their residues have mixed signs at up to 11 of the 19 derivative levels, and the realized DC gain is not positive at up to 5 levels. The D- and E-type inverses are then unstable there, and Proposition 8 cannot certify the others. With the sign and DC constraints, all 19 levels are certified, and the saving drops to 20 to 30 % ($K = 24$ against 30 at 3×10^{-6} , $K = 28$ against 39 at 2×10^{-7}). The analytic rule thus does not give the smallest bank. It trades a fifth to a quarter of the states for a closed-form design that needs no optimization, was validated on a sealed holdout, and keeps the residue signs by construction.

8 Applications

8.1 Online order tracking

Consider a fractional element, such as an ultracapacitor or a viscoelastic material, excited by a known input x while its memory exponent drifts. The task is to track the order from the noisy output

$$y_n = \left({}^T\Delta^\alpha x\right)_n + v_n, \quad T \in \{A, B\}, \quad (30)$$

with a random-walk prior $\alpha_{n+1} = \alpha_n + w_n$. Unscented fractional-order Kalman filters have been used to estimate a variable order jointly with the state [11]. The bank contributes the measurement model and its derivative at $O(K)$ cost for either type, and its structure decides which estimator is natural.

A-type.

With the order in the output weights, the states s_n and $s_d[n]$ of (17) are driven by the known input alone. Hence

$$y_n = \varphi_n(\alpha_n) + v_n, \quad \varphi_n(a) = \hat{g}_0(a)x_n + c_d(a)s_d[n] + c(a)^\top s_n, \quad (31)$$

a memoryless measurement of the current order, whose regressors are shared by all orders. An extended Kalman filter (EKF) needs φ_n and φ'_n , which cost $O(K)$. Because the regressors do not depend on the order, the exact likelihood of G order hypotheses costs $O(GK)$, and a grid (point-mass) Bayes filter becomes affordable.

B-type.

With the order in the input weights, the state of (18) carries the order history. The filter state is (s, s_d, α) , of dimension $K + 2$; the model and its Jacobians cost $O(K)$ and the covariance update $O(K^2)$. In the literal GL form every past order enters the output, so a B-type filter would have to carry the whole order history.

Experiment.

We used sample units ($T_s = 1$, so that $\hat{g}_0 = 1$ and the order enters only through the memory), $n = 2000$, and an order that is constant at 0.3, ramps to 0.7, jumps to 0.2 and then oscillates (Fig. 6). The input was a known AR(1) process with pole 0.98. The data came from the literal GL definitions, with white noise at 20, 40 and 60 dB SNR, in 50 Monte Carlo runs. One rule-designed bank ($\varepsilon = 10^{-5}$, $|\alpha| \leq 0.95$, $K = 40$, coefficients interpolated by cubic splines on a grid of spacing 0.0025) served three estimators:

- an A-type grid filter on 381 orders, with a Gaussian random walk mixed with a uniform switch of probability 10^{-3} , and a likelihood averaged over each grid cell;
- an A-type EKF with an iterated measurement update;
- a B-type EKF on the augmented state.

Table 7 Order tracking (Sect. 8.1): RMSE of the order over $0.2 \leq t < 4$ s, median [90th percentile] over 50 runs. “Wrong” marks data of the other type. Costs are multiplications per sample ($K = 40$, $G = 381$, $n = 2000$). The B-type window $W = 20$ gave the same values as $W = 5$

Estimator (cost)	Data	20 dB	40 dB	60 dB
A-type grid filter ($G(K + 2) \approx 16\,000$)	A	0.018 [0.025]	0.0040 [0.005]	0.0013 [0.002]
	B, wrong	0.029 [0.042]	0.059 [0.111]	0.140 [0.272]
A-type EKF ($\approx 3(K + 2)$ per iteration)	A	0.026 [0.043]	0.0083 [0.110]	0.017 [0.154]
	B, wrong	0.037 [0.056]	0.099 [0.302]	0.213 [0.370]
B-type EKF ($\approx 4(K + 2)^2 \approx 7100$)	B	0.027 [0.040]	0.015 [0.017]	0.012 [0.013]
	A, wrong	0.032 [0.053]	0.054 [0.088]	0.238 [0.478]
<i>Unscented filters on the GL sum (UFOKF-type, our implementation)</i>				
A-type, memory $L = 100$ ($\approx 6L = 600$)	A	0.036 [0.052]	0.138 [0.236]	0.244 [0.327]
A-type, memory $L = 500$ (≈ 3000)	A	0.028 [0.043]	0.026 [0.220]	0.081 [0.282]
A-type, full memory ($\approx 3n$ on average)	A	0.027 [0.043]	0.016 [0.115]	0.039 [0.162]
A-type, same UKF on the bank ($\approx 6(K + 2)$)	A	0.027 [0.043]	0.016 [0.115]	0.039 [0.162]
B-type, window $W = 5$ ($\approx n + (2W + 1)W^2$)	B	0.028 [0.039]	0.014 [0.017]	0.012 [0.012]

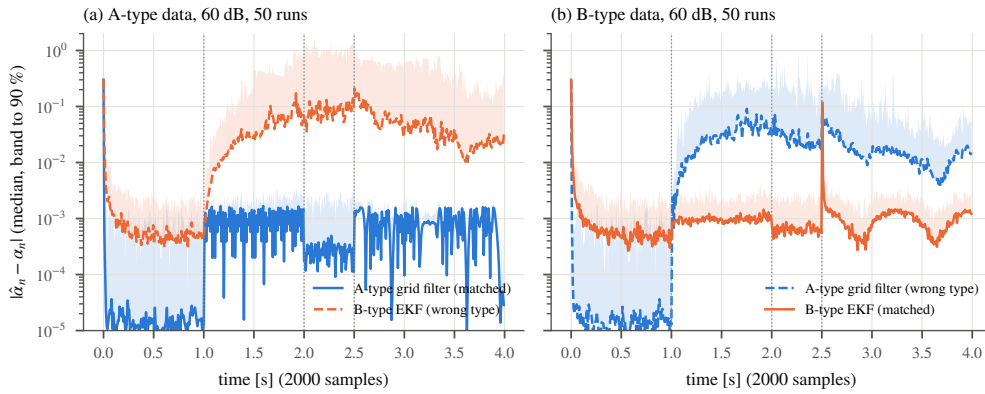


Fig. 6 Order tracking at 60 dB: absolute error of the order estimate (median over 50 runs, band to the 90th percentile). Solid: filter of the type that generated the data; dashed: filter of the other type. Dotted lines mark the start and end of the ramp and the jump

Both EKFs use the random-walk variance 10^{-5} and inflate the order variance by 5×10^{-4} when the normalized innovation exceeds 16, which detects jumps. Every estimator was also run on data of the other type.

Results.

Table 7 and Fig. 6 show two regimes.

- *Matched type.* The error decreases with the SNR. At 60 dB the A-type grid filter reaches a median RMSE of 1.3×10^{-3} and the B-type EKF 1.2×10^{-2} . The A-type EKF on the bank reproduced the same EKF on the literal GL weights to 9×10^{-6} , at about $3(K + 2)$ instead of $3n$ multiplications per iteration.
- *Wrong type.* The error increases with the SNR, to 0.14–0.24 at 60 dB, which is 10 to 100 times the matched error. In the half second after the jump, the filters of the

wrong type were off by 0.13 to 0.30. A wrong definition biases the estimate, and more data does not remove the bias. While the order is still constant ($t < 1$ s), the two definitions coincide and both filters are accurate; the bias appears as soon as the order varies (Fig. 6).

The choice of definition is therefore a modelling decision for inference as well, and the bank lets both models be fitted and compared at the same cost. The A-type EKF has a heavy tail at high SNR (90th percentile 0.11 to 0.15). At the jump, the likelihood of a single sample has several maxima, and the iterated update can lock onto the wrong one. The grid filter avoids this at about 16 000 multiplications per sample, still independent of n .

Unscented filters on the GL sum.

Unscented fractional-order Kalman filters keep the order in the filter state and evaluate the GL memory directly [11]. We implemented filters of this kind for (30), with the same random walk and jump gating; the cited filter also estimates the states of a fractional state-space model, which (30) does not have. For the A-type, a scalar UKF evaluates the GL sum over the last L samples. For the B-type the current order is not seen in y_n when $T_s = 1$, so the state holds the last W orders; older orders are frozen at their estimates, and their contributions to later outputs are accumulated by the literal B-type sum. Table 7 (lower part) shows three results.

- With a truncated memory the omitted tail biases the estimate: at 60 dB the median RMSE is 0.24 for $L = 100$ and 0.081 for $L = 500$, against 0.039 with the full memory.
- With the full memory the GL filter and the same filter on the bank agreed to 10^{-4} in 144 of 150 runs (median difference 3×10^{-6}); in the other six they separated briefly at the jump but gave the same RMSE to three digits. The bank thus gives the full-memory filter at $6(K + 2)$ instead of about $3n$ multiplications per sample. Both are local filters, and at 40 and 60 dB the grid filter, which is affordable only because the bank's regressors do not depend on the order, is 4 to 30 times more accurate.
- For the B-type, the windowed UKF matched the bank's EKF (0.012 at 60 dB), but it keeps and updates the whole input history, $O(n)$ operations and memory per sample, whereas the bank's state has $K + 2$ entries.

With a physical sampling time the gain $T_s^{-\alpha}$ adds direct information about the current order and makes the problem easier.

8.2 Measured data: a lithium-ion cell

Fractional elements are common in battery models [56], and variable-order versions let the order depend on the state of charge [57–59]. The question of this paper applies to such models. Once the order moves, a model identified with one definition and executed by a realization of the other is a different model. We tested how much this matters on public measurements of a Panasonic 18650PF cell [60]:

- four drive cycles (UDDS, LA92, US06, HWFET) at each of five chamber temperatures from 25 to -20°C ;

- two warming runs of four mixed cycles each, during which the cell temperature rose from 10 to 24–27 °C and from –20 to 12–17 °C.

Model.

The signals are the terminal voltage v , the current i (positive for charge), the state of charge z from coulomb counting and the open-circuit voltage U of the C/20 test at 25 °C. The model is

$$v_n - U(z_n) = c(z_n) + R_0(z_n) i_n + R_1(z_n) (\psi_\tau i)_n + \kappa(z_n) \left({}^T\Delta^{-\beta(z)} i \right)_n, \quad T \in \{A, B\}, \quad (32)$$

at $T_s = 1$ s, using means over 1 s of the 10 Hz log.

- ψ_τ is a first-order lag of unit DC gain.
- The gains c, R_0, R_1, κ are piecewise linear in z , with knots 0.1 apart.
- The order of the fractional integral is piecewise linear in z on six knots, so it changes at every sample.

In the A-type the order of the present sample weights the whole past current. In the B-type each current sample keeps the order of the moment it flowed. As baselines, the same model was fitted with a constant order (where A and B coincide), and with one or two lags in place of the fractional term (1RC, 2RC).

For fixed time constants and orders, the gains follow from least squares. The time constant (40 values from 1 to 5000 s) and the order knots (0.02 to 0.95) were found by coordinate search. One rule-designed bank served all orders and both types ($\varepsilon = 10^{-4}$, $R = 25\,000$, $\beta \in [0.02, 0.95]$, $K = 29$). The A-type columns of all 94 grid orders come from one state, and a candidate order path costs $O(nK)$ for either type. The whole study ran in 15 minutes on four cores.

Protocol.

Within each group (one chamber temperature, or one warming run), every model was fitted jointly to three cycles. It then simulated the fourth from its current alone, with every parameter, the order included, read from the z of the held-out record (leave one cycle out). We also ran each fitted VO model with the operator of the other type. This is what happens when a model identified with one definition is executed by a realization of the other (Sect. 4).

Fits to a single cycle did not transfer: over 84 single-cycle runs, the RC+FO models went from 12 mV in sample to 92–101 mV on the other cycles, and the 1RC model from 17 to 46 mV. The slow terms are nearly collinear with z itself, the integral of the current, and another cycle breaks that balance. Joint fits to three cycles bring the held-out errors of the RC+FO models down to the level of the RC models.

Results.

Table 8 and Fig. 7 give three results.

- *The definition matters on measured data.* Run with the A-type operator, the B-type models were worse in all 28 folds, by a median factor of 4.4 (at least 1.06). Run with

Table 8 Battery data (Sect. 8.2): RMSE in mV of the simulated voltage on the held-out cycle, mean over the four leave-one-cycle-out folds of each group. The last two columns run the fitted VO models with the operator of the other type. Bold: lowest error of the row

Group	1RC	2RC	RC + fractional integral			other type	
			constant	VO-A	VO-B	A as B	B as A
25 °C	20.2	18.6	16.3	15.8	14.7	94	46
10 °C	27.2	27.0	25.6	37.7	26.7	44	98
0 °C	36.1	35.0	36.2	36.8	36.3	49	92
−10 °C	60.6	62.7	66.7	66.7	66.9	104	328
−20 °C	111.1	116.4	122.3	122.8	128.5	239	1295
10 °C, warming	14.7	13.5	15.1	14.9	14.6	23	80
−20 °C, warming	34.4	33.1	36.8	38.1	40.4	94	109

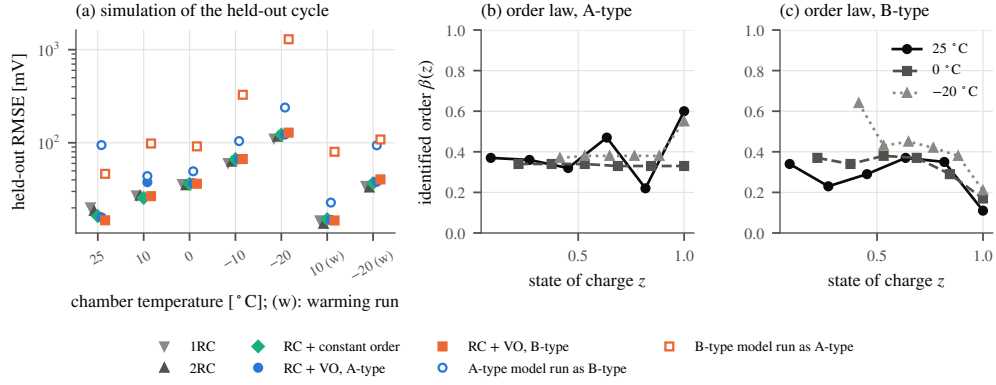


Fig. 7 Battery data. (a) Held-out RMSE (Table 8); hollow markers: the fitted VO models run with the operator of the other type. (b, c) Order laws $\beta(z)$ fitted to all four cycles of a temperature with the A- and B-type operators

the B-type operator, the A-type models were worse in 23 of 28 folds, by a median factor of 2.1. The A-type laws are almost constant, and for a constant order the two types coincide; this is consistent with the smaller effect.

- *The fitted order laws depend on the definition* (Fig. 7b, c). At the five constant temperatures, the B-type order drops at the start of the discharge, to 0.11–0.21 at $z = 1$, against 0.23–0.64 at lower states of charge. Away from $z = 1$, the A-type laws stay within 0.33–0.43, except at 25 °C (0.22–0.47). At $z = 1$ the A-type order is 0.33–0.60, above the B-type order at every temperature. An order law reported for a VO battery model is therefore meaningful only together with its definition.
- *A varying order helped only at the highest temperature.* At 25 °C the B-type model had the lowest held-out error (14.7 mV, against 16.3 for a constant order and 18.6 for 2RC) and was best in three of four folds. At 10 °C it was within 1.1 mV of the constant order. At 0 °C and below, and in both warming runs, the best model was an RC model. At 0 °C and below every model had held-out errors between 35 and 128.5 mV.

The low-temperature errors are not a matter of the order. We linearized each model at the state of charge of the impedance sweeps in the same data set and compared it with the measured impedance between 1.4 mHz and 0.1 Hz. Every model's impedance magnitude was about 0.9 of the measured one at 25 and 10 °C, 0.7 at 0 °C, 0.6 at −10 °C and 0.4 at −20 °C (medians over 8 to 12 sweeps). The models fitted to the drive cycles thus have a lower impedance than the cell under small-signal excitation, increasingly so at low temperature. This is consistent with a charge-transfer resistance that falls with the current amplitude, and with self-heating during the cycles, and neither effect is in a model that is linear in the current.

The experiment therefore does not show that a variable order improves battery models in general. It shows that, on measured data, the definition is a modelling choice with a large effect, and that one bank fits and runs both definitions at the same cost.

8.3 Streaming multifractional Brownian motion

Multifractional Brownian motion lets the Hurst exponent vary in time [61], and its definitions are in general not equivalent [62]. The Riemann–Liouville form (RL-mBm) [63, 64] is a VO integral of white noise of order $H(t) + \frac{1}{2}$, with the current order applied to the whole past. It is thus an A-type integral. Applying the order at the time each increment enters gives the B-type variant. Up to the discretization of the kernel and a per-increment factor $\sqrt{2H_j} \Gamma(H_j + \frac{1}{2})$, this is the memory-multifractional Brownian motion (MMFBM) of Wang et al. [65], $X(t) = \int_0^t \sqrt{\alpha(s)} (t-s)^{(\alpha(s)-1)/2} dB(s)$ with $\alpha = 2H$. The input-scheduled bank applies that factor as a gain on each input sample. Other multifractional constructions treat a varying H differently [45, 66–68]; they are compared in Supplementary Sect. S5. VO operators have been used to synthesize multifractional Gaussian noise before [69], and simulation methods for mBm include [70].

In sample units, the discrete processes are

$$X_n^A = \sum_r g_r(-a_n) \xi_{n-r}, \quad X_n^B = \sum_j g_{n-j}(-a_j) \xi_j, \quad a_n = H_n + \frac{1}{2},$$

with i.i.d. standard normal ξ . For $H > \frac{1}{2}$ the order exceeds 1, outside the range of the bank. An integer integration can be split off exactly, but only in the right place:

Proposition 11 (Integer split) *Let Σ denote the cumulative sum. Then*

$${}^A\Delta^{-a}\xi = {}^A\Delta^{-(a-1)}(\Sigma\xi), \quad {}^B\Delta^{-a}\xi = \Sigma({}^B\Delta^{-(a-1)}\xi).$$

Proof $(1 - z^{-1})^{-a} = (1 - z^{-1})^{-(a-1)}(1 - z^{-1})^{-1}$ gives $w_m(-a) = \sum_{i=0}^m w_i(-(a-1))$ for every fixed a . For the A-type, apply this at the fixed order a_n of output sample n and exchange the sums. For the B-type, sample j keeps its order a_j :

$$\sum_{j \leq n} \sum_{i \leq n-j} w_i(-(a_j-1)) \xi_j = \sum_{m \leq n} \sum_{j \leq m} w_{m-j}(-(a_j-1)) \xi_j.$$

The right-hand side is the cumulative sum of ${}^B\Delta^{-(a-1)}\xi$. □

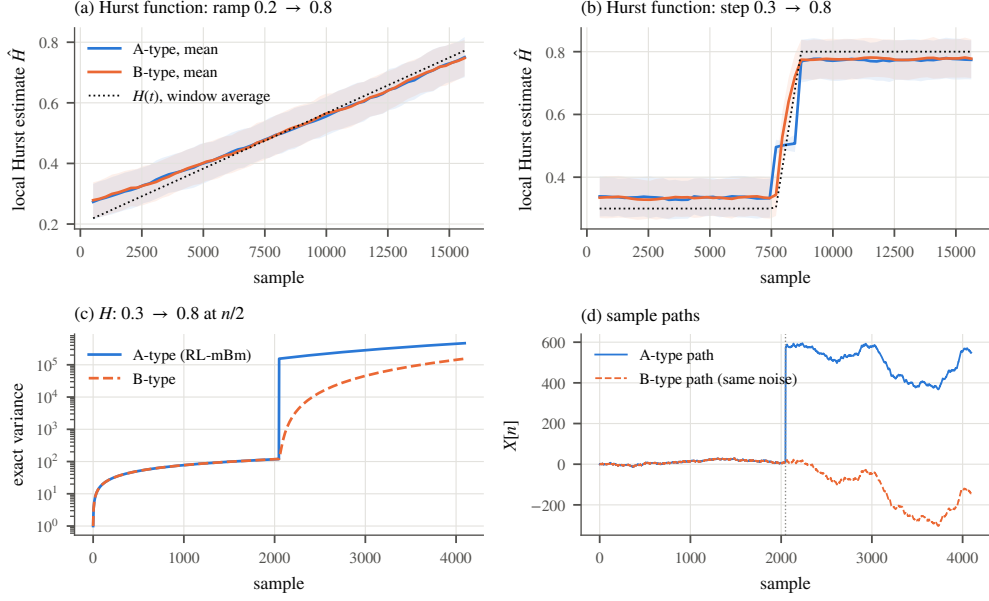


Fig. 8 Streaming mBm. (a, b) Local Hurst estimates (mean $\pm$ one standard deviation over 400 paths) for a ramp and a step of H . (c) Exact variance of the A-type (RL-mBm) and B-type processes for a jump of H at $n/2$. (d) Sample paths driven by the same noise

The reversed orders are not exact: the relative errors are 0.1 to 1.3 in our tests. With the split, one bank of order $\frac{1}{2} - H$, which satisfies $|\frac{1}{2} - H| < \frac{1}{2}$, covers all of $0 < H < 1$. It crosses $H = \frac{1}{2}$ seamlessly because the order 0 is exact in the bank.

Results.

The bank was designed by the rule for $\varepsilon = 10^{-4}$ and $0.05 \leq H \leq 0.95$ ($K = 22$ to 26 for $n = 1024$ to 16384). Paths agree with the literal GL processes driven by the same noise to 1.9×10^{-6} (A) and 2.4×10^{-6} (B). Computed exactly from the weight matrix, the variance agrees with the exact formula to 3.1×10^{-5} and the covariance to 1.7×10^{-6} . The local Hurst exponent, estimated by generalized quadratic variations [71, 72], follows a varying $H(t)$ with an RMSE of the mean of 0.014 to 0.036 for both types (Fig. 8). The bank needs 47 to 55 multiplications per sample, independent of n , against $n/2$ on average for the direct GL sum. Further results, including the fixed-point generators, are in Supplementary Sect. S5.

The two types differ sharply at a jump of H from 0.3 to 0.8. The A-type variance jumps from 119 to 1.5×10^5 , and the path is discontinuous: the squared increment at the jump is 1.3×10^5 times the typical one. The B-type is continuous, in line with the behaviour reported for MMFBM and incremental mBm [65, 68]. Which definition to use is therefore a modelling decision with visible consequences [65], and the bank makes either choice available at the same cost.

Algorithm 1 One sample of the bank, all four types. In hardware the coefficient row is a ROM word addressed by the order index (mirrored for D and E), so an order or type change costs nothing

Require: type $T \in \{A, B, D, E\}$, order α_n , input x_n ; states $s_0, \dots, s_{K-1}, s_d$

```

1:  $a \leftarrow \alpha_n$  if  $T \in \{A, B\}$ , else  $a \leftarrow -\alpha_n$ ; read  $(c_d, c_0, \dots, c_{K-1})$  for order  $a$ 
2: if  $T \in \{A, D\}$  then ▷ output schedule
3:    $\eta \leftarrow c_d s_d + \sum_k c_k s_k$ 
4: else ▷ input schedule
5:    $\eta \leftarrow s_d + \sum_k s_k$ 
6: end if
7: if  $T \in \{A, B\}$  then  $v \leftarrow x_n$ ;  $y_n \leftarrow v + \eta$ 
8: else  $v \leftarrow x_n - \eta$ ;  $y_n \leftarrow v$  ▷ no divider in sample units
9: end if
10: if  $T \in \{A, D\}$  then  $s_k \leftarrow \theta_k s_k + v$ ;  $s_d \leftarrow v$ 
11: else  $s_k \leftarrow \theta_k s_k + c_k v$ ;  $s_d \leftarrow c_d v$ 
12: end if

```

9 Fixed-point realization

The design point is $R = 4000$, $\varepsilon = 10^{-4}$ and $|\alpha| \leq 0.95$, giving $K = 32$ plus the delay mode. Orders come from a grid of $P = 1024$ levels, $\alpha_i = 0.95(2i/(P-1) - 1)$, so $\Delta\alpha = 1.9 \times 10^{-3}$.

The core runs in sample units ($T_s = 1$), where $\hat{g}_0 = 1$ for every order, so the inversion (21) needs no divider: $v_n = x_n - \eta_n$. Physical units follow from one gain $T_s^{-\alpha_n}$, applied at the output for the A- and E-types and at the input for the B- and D-types. Signals are W_S -bit two's complement numbers, states have $W_S + 13$ bits with a power-of-two scaling per mode, and coefficients have a W_M -bit mantissa with a 7-bit shift. Algorithm 1 gives one sample in exact arithmetic; the bit-accurate golden model and the RTL use the same sequence.

Two effects set the word lengths (Supplementary Sect. S4).

- *Dynamic range.* Under switching from an integral to a derivative order, the D- and E-types reach 820 times the frozen-order ℓ^1 bound on the output. This is a property of the recursive definitions, not of the bank. We therefore size two envelopes: forward types only (A/B, 14 integer bits) and all four types (A/B/D/E, 24 integer bits).
- *Quantization and inverse stability.* Quantization can undo the DC floor. Coefficients and pole steps are dyadic rationals, so the signs of $\hat{H}_q(\pm 1)$ of every quantized ROM row were evaluated exactly in rational arithmetic. With the floating-point floor alone, 64 (A/B) and 51 (A/B/D/E) of the 512 positive-order rows had $\hat{H}_q(1) \leq 0$; the worst inverse pole was at $1 + 1.9 \times 10^{-9}$. A quantization-aware floor, which raises the delay mantissa in LSB steps until $\hat{H}_q(1) \geq \hat{H}(1)$, brings both counts to zero.

A configuration was accepted if every type stayed within 0.1ε of the floating-point bank on a stress set of order profiles and inputs. The smallest accepted configurations are $W_S/W_{ST}/W_M = 36/49/16$ (A/B) and $48/61/25$ (A/B/D/E).

Table 9 The two configurations on a Zynq-7020: resources and $f_{\max}$ of the core (Vivado, out of context), and bit-exact parity on the board (PYNQ-Z1). Sample rates are $f/135$

Types	$W_S/W_{ST}/W_M$	core, out of context					board		
		LUT	FF	DSP	BRAM	$f_{\max}/\text{MHz}$	clock/MHz	vectors (mism.)	MS/s
A/B	36/49/16	3 214	1 882	11	46	44.3	40.0	31 744 (0)	0.296
A/B/D/E	48/61/25	4 278	2 337	15	64	39.3	34.5	62 464 (0)	0.255

RTL, parity and resources.

The core (`vo_bank_core.v`) is a non-pipelined, time-multiplexed state machine with synchronous coefficient, pole and gain ROMs. It produces one sample every $4K + 7 = 135$ clock cycles. Order index and type are inputs with every sample. An order change only changes a ROM address: there is no reload and no extra cycle.

The test vectors cover constant, piecewise and per-sample orders, sign-changing switches, full-scale inputs, and type changes at run time. The type changes exercise the datapath; a type change is not itself a VO operation. Icarus Verilog and Vivado simulation matched the golden model bit for bit on all 94 208 vectors (31 744 for A/B, 62 464 for A/B/D/E). Table 9 gives the Vivado results for a Zynq-7020 (xc7z020clg400-1, Vivado 2026.1, out-of-context implementation of the core).

Board measurements.

Each configuration ran on a PYNQ-Z1 board behind an AXI4-Lite wrapper driven by the on-chip processor, with one reset and then all vectors in one stream. Both were bit-exact: the outputs were byte-identical to the simulation, with 0 mismatches (Table 9). A cycle counter confirmed 135 cycles from input acceptance to output and between back-to-back samples. The critical path runs from the coefficient ROM through the multipliers and the state adder in one cycle (43 logic levels for A/B/D/E); pipelining it would raise the clock.

For the long run, the D-type ran at the worst quantized row above ($\alpha = -0.855$, whose inverse is the derivative row of order 0.855) on $2^{24} \approx 1.7 \times 10^7$ samples of an on-chip 32-bit LFSR input with $|x| \leq 0.25$. The board kept, for every block of 2^{16} outputs, the maximum $|y|$, the sum of y^2 and a CRC-32. All 256 block records were identical to the golden model over the whole run. The golden model saw no overflow (peak state 51.3 of 61 bits, peak signal 29.4 of 48 bits), and the largest output, $|y| = 85.7$, was far below full scale (2^{24}). The block RMS ranged from 2.1 to 79, as a fractional integral of order 0.855 of white noise wanders. This run shows bit-exactness and boundedness in hardware; it cannot by itself separate the two floors, because the floating-point floor's unstable pole would grow by only a factor of 1.03 over 2^{24} samples. Stability rests on the exact sign test above. The measurements were made with the new core; none is taken from [23].

10 Discussion and limitations

Scope of the guarantees.

Proposition 2 and Corollary 3 hold for every order sequence. Proposition 8 is a statement about each *frozen* order. The recursive types under switching form a switched linear system $F - b c(\alpha_n)^\top / \hat{g}_0$, whose stability under arbitrary switching is not implied by the stability of each member. In all our tests the switched D- and E-types stayed bounded, within the stress envelope of Sect. 9, but we have no proof for arbitrary switching.

Output versus weight error.

Theorem 5 turns the weight guarantee into an operator guarantee: for the A- and B-types, the worst-case relative output error over all inputs, in ℓ^1 and ℓ^∞ , is at most ε_R . Particular inputs can still show larger relative errors, because a derivative of a smooth input is much smaller than the input. The relative output error then exceeded ε_R by up to a factor of seven for the bank of Sect. 7.5, and by up to six for the D- and E-types at the design point of Sect. 9. For the recursive integrals the operator error itself is amplified, by up to $\kappa_\infty \approx 2N^a / \Gamma(1+a)$ (Remark 2); a guaranteed design for them must use $\varepsilon / \kappa_\infty$, which adds about $K \ln \kappa_\infty / \ln(1/\varepsilon)$ modes.

Size of a common pole set.

Theorem 9 gives an upper bound on K , not a lower one. We do not know how much a pole set shared by all orders must exceed the best pole set for a single order. In the comparison of Sect. 7.8, per-order grids reach the same constant-order accuracy with about two thirds of the states, but they cannot be switched without a definitional error.

Order range.

The bank covers $-1 < \alpha < 1$. Larger orders follow from integer splits such as Proposition 11, which must be placed according to the type.

Fixed point.

We used a single Q format per signal. Block floating point could reduce the D/E word lengths, which are dominated by the switching envelope.

Hardware.

The core is not pipelined and takes $4K + 7$ cycles per sample at about 40 MHz on a Zynq-7020. A pipelined version could approach K plus a few cycles per sample and a higher clock.

Estimation.

The tracking experiment assumes a known input and a known noise level. Joint estimation of the state or the input, as in dual and triple estimation with unscented fractional-order Kalman filters [11], can use the same $O(K)$ model but was not tested here.

Battery data.

The battery model of Sect. 8.2 is linear in the current, uses one open-circuit voltage curve for all temperatures and has no hysteresis. It is meant to compare definitions under one model, not to compete with battery models. Below 0 °C its errors are dominated by effects outside it. The fitted order laws are not unique either: the lag and the fractional term can trade time scales, and other model structures may give other laws. That the same data give different laws under the A- and B-types does not depend on these choices, because both types were fitted under the same structure.

Literature search.

The review of Sect. 1.1 used Crossref and arXiv. [TODO: repeat the documented queries in Scopus and Web of Science before submission.]

11 Conclusion

For a finite-dimensional realization of a variable-order operator, the definition it implements is a property of its structure. If the poles move with the order, no state map can make a switched realization exact for the A- or B-type. If the poles are fixed and the order sits in the output or input weights, the realization is the A- or B-type for every order sequence, up to the frozen-order quadrature error. Inverting the same bank gives the D- and E-types with exact duality. The Beta-integral representation makes this exact in discrete time. The weight error bounds the realized operators as a whole: for the forward types the relative operator error never exceeds it, and for the recursive integrals it is amplified by a computable condition number. Nothing in the construction is specific to Grünwald–Letnikov weights: every kernel family that is completely monotone in the lag, such as distributed orders with a time-varying density, is realized in the same way, with a common pole set whose size grows like $\ln(1/\varepsilon) \ln(R/\varepsilon)$. An analytic rule sizes the bank, and a DC floor, tested by an exact sign, keeps every inverse stable, also after quantization. In hardware the order and the type become a ROM address that can change at every sample. Order tracking, measured battery data and streaming multifractional Brownian motion show that the choice between the A- and B-type is a modelling decision with visible consequences, in inference as in synthesis, and the bank makes either choice available at the same cost.

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Declarations

- **Funding:** [TODO.]
- **Competing interests:** The authors declare no competing interests. [TODO: confirm.]
- **Data availability:** All results reported in this paper are produced by the scripts in the accompanying repository, with fixed seeds; the result files are included. The battery measurements of Sect. 8.2 are public (Mendeley Data, doi:10.17632/wykht8y7tg.1, CC BY 4.0 [60]); a script in the repository downloads them and checks their SHA-256 sums.

- **Code availability:** [TODO: public repository URL or archive DOI.] The repository contains the Python library, the experiments, the Verilog core, the testbench, the board wrapper and host scripts, and the bitstreams with their SHA-256 sums.
- **Author contributions:** [TODO.]
- **Relation to other work:** A constant-order FPGA approximation by the same authors is under review at AEÜ – International Journal of Electronics and Communications [23]. The present paper uses different structures (order-independent poles), a different design rule (analytic, fit-free) and a new core. It reuses no figure, table, text or measurement from that work.

Appendix A Proof of Theorem 6

Write $\mathcal{C}_i = [b_i, F_i b_i, \dots, F_i^{S-1} b_i]$ and $\mathcal{O}_i = [c_i, F_i^\top c_i, \dots, (F_i^\top)^{S-1} c_i]^\top$, both invertible by minimality [73].

(a) For $n \geq m$ the switched output is $d_2 v_n + c_2^\top F_2^{n-m} \Phi x_1[m] + c_2^\top \sum_{m \leq j < n} F_2^{n-1-j} b_2 v_j$. The response of Σ_2 to the whole input differs from it only by $\Phi x_1[m]$ being replaced by $x_2[m] = \sum_{j < m} F_2^{m-1-j} b_2 v_j$. Equality for all inputs and all $n \geq m$ means $c_2^\top F_2^k (\Phi x_1[m] - x_2[m]) = 0$ for all $k \geq 0$. By observability this holds if and only if $\Phi x_1[m] = x_2[m]$ for every input history, that is, if and only if $\Phi F_1^k b_1 = F_2^k b_2$ for $k = 0, \dots, m-1$. Since $m \geq S+1$, this gives $\Phi \mathcal{C}_1 = \mathcal{C}_2$, so Φ is invertible. It also gives $(\Phi F_1 - F_2 \Phi) F_1^k b_1 = \Phi F_1^{k+1} b_1 - F_2^{k+1} b_2 = 0$ for $k = 0, \dots, S-1$, hence $(\Phi F_1 - F_2 \Phi) \mathcal{C}_1 = 0$ and $F_2 = \Phi F_1 \Phi^{-1}$. Conversely, if $F_2 = \Phi F_1 \Phi^{-1}$ and $b_2 = \Phi b_1$, then $\Phi F_1^k b_1 = F_2^k b_2$ for all k .

(b) The B-type behaviour requires that the stored state continue as in Σ_1 : $c_2^\top F_2^k \Phi x = c_1^\top F_1^k x$ for all $k \geq 0$ and every reachable x . By controllability every x is reachable within $S \leq m-1$ samples, so $\mathcal{O}_2 \Phi = \mathcal{O}_1$. Then Φ is invertible, $c_2^\top = c_1^\top \Phi^{-1}$, and $\mathcal{O}_2 (F_2 \Phi - \Phi F_1) = 0$ follows in the same way, so $F_2 = \Phi F_1 \Phi^{-1}$. The converse is immediate. $\square$

Appendix B Proof of Proposition 8

N has degree $K+1$. If $c_d = 0$, then $z = 0$ is a root of N ; it cancels the delay pole and lies inside the disc. We give the argument for $c_d \neq 0$; the case $c_d = 0$ is the same with the pole at 0 removed.

(i) All residues are positive, so $\hat{H}$ decreases from $+\infty$ to $-\infty$ between consecutive poles of $\{0, \theta_0, \dots, \theta_{K-1}\}$. This gives K roots in $(0, \max_k \theta_k)$. On $(\max_k \theta_k, \infty)$, $\hat{H}$ decreases from $+\infty$ to $g_0 > 0$, so there is no root there. On $(-\infty, 0)$, $\hat{H}$ decreases from $g_0 > 0$ to $-\infty$, so there is exactly one root z_* . It lies in $(-1, 0)$ if and only if $\hat{H}(-1) > 0$.

(ii) Between consecutive θ_k the negative residues make $\hat{H}$ rise from $-\infty$ to $+\infty$, which gives $K-1$ roots. For $z > \max_k \theta_k$,

$$z \hat{H}(z) = g_0 z + c_d + \sum_k c_k (1 + \theta_k / (z - \theta_k))$$

is strictly increasing from $-\infty$ to $+\infty$. Hence there is exactly one root there, and it is smaller than 1 if and only if $\hat{H}(1) > 0$. If $c_d < 0$, then $z\hat{H}(z)$ rises from $c_d < 0$ at $z = 0$ to $+\infty$ at $\min_k \theta_k^-$, giving the last root in $(0, \min_k \theta_k)$. If $c_d > 0$, then $z\hat{H}(z)$ rises from $-\infty$ at $z \rightarrow -\infty$ to $c_d > 0$ at $z = 0$. Counting shows there is exactly one root $z_* < 0$, with $z\hat{H}(z) < 0$ for $z < z_*$. Hence $z_* > -1$ if and only if $-\hat{H}(-1) < 0$, that is, $\hat{H}(-1) > 0$. In all cases the $K + 1$ roots are real, so no complex root can lie outside the disc.

(iii) Integral bank, order $-a$ with $0 < a < 1$. Since $\hat{g}_1 = c_d + \sum_k c_k$ and $\hat{g}_2 = \sum_k c_k \theta_k$,

$$\hat{H}(-1) = g_0 - \hat{g}_1 + \sum_k \frac{c_k \theta_k}{1 + \theta_k} \geq g_0 - \hat{g}_1 + \frac{1}{2} \hat{g}_2.$$

With $g_1 = ag_0$, $g_2 = \frac{1}{2}a(1+a)g_0$ and relative errors at most $\delta = \frac{1}{3}$, the right-hand side is at least $g_0(1-a)(6-a)/6 > 0$. Derivative bank, order a : here $\sum_k c_k \theta_k / (1 + \theta_k) \geq \hat{g}_2$ because $c_k < 0$. Hence $\hat{H}(-1) \geq g_0 - \hat{g}_1 + \hat{g}_2$. With $g_1 = -ag_0$ and $g_2 = -\frac{1}{2}a(1-a)g_0$, this is at least $g_0(1 + \frac{2}{3}a^2) > 0$. $\square$

Appendix C Proof of Theorem 5

(a) Row n of E_A is $(e_n(\alpha_n), \dots, e_1(\alpha_n), 0, \dots, 0)$. It contains a single order, so its absolute sum is at most ρ_{row} . Column j of E_A contains $e_{n-j}(\alpha_n)$ for $n > j$, one entry per lag, each at most $\sup_a |e_{n-j}(a)|$, so its absolute sum is at most ρ_{col} . In E_B column j contains the single order α_j , and row n contains one entry per lag. This gives the four bounds. The ℓ^2 bound follows from $\|E\|_2^2 \leq \|E\|_1 \|E\|_\infty$. For a finite $\mathcal{S}$, let a^* maximize $\sum_r |e_r(a)|$ and a_r maximize $|e_r(a)|$. The constant sequence a^* attains ρ_{row} in the last row of E_A and in the first column of E_B . The sequence $\alpha_n = a_n$ attains ρ_{col} in the first column of E_A , and the reversed sequence $\alpha_n = a_{N-1-n}$ attains it in the last row of E_B .

(b) The off-diagonal entries satisfy $|(E_T)_{n,j}| \leq \varepsilon_R |(W_T)_{n,j}|$ because $n-j < N \leq R$, and the diagonal of E_T vanishes. For any x , $|E_T x| \leq |E_T| |x| \leq \varepsilon_R |W_T| |x|$ entrywise, so $\|E_T x\|_p \leq \varepsilon_R \|W_T\|_p \|x\|_p$. For $p \in \{1, \infty\}$ the induced norm depends only on the absolute values of the entries.

(c) Since $\hat{W}_T = \hat{W}_F(-\alpha)^{-1}$ and $W_T = W_F(-\alpha)^{-1}$,

$$\hat{W}_T - W_T = \hat{W}_T (W_F(-\alpha) - \hat{W}_F(-\alpha)) W_T = -\hat{W}_T E_F^- W_T.$$

Hence $\|E_T\|_p \leq \|\hat{W}_T\|_p \|E_F^-\|_p \|W_T\|_p \leq (\|W_T\|_p + \|E_T\|_p) q$, which rearranges to the first bound. Writing $\|W_T\|_p \leq \|\hat{W}_T\|_p + \|E_T\|_p$ instead gives the second. If all orders are negative, $\hat{W}_F(-\alpha)$ is a derivative bank at every sample: its diagonal $T_s^{\alpha_n}$ is positive, and its off-diagonal entries $\hat{g}_r(-\alpha_n)$, $r \geq 1$, are nonpositive, for $r \geq 2$ because all $c_k < 0$ and for $r = 1$ by assumption. (The DC floor changes $\hat{g}_1$ only by about the DC quadrature error; in the 200 discrete-time holdout designs, $\hat{g}_1 < 0$ at all 4294 checked derivative orders.) Write $\hat{W}_F(-\alpha) = \Delta - L$ with Δ diagonal positive and $L \geq 0$ strictly lower triangular. Then $\hat{W}_T = \sum_{k=0}^{N-1} (\Delta^{-1} L)^k \Delta^{-1} \geq 0$, and for a nonnegative matrix $\|M\|_\infty = \|M \mathbf{1}\|_\infty$. $\square$

Appendix D Proof of Theorem 9 and Lemma 10

Part (a) is stated after the theorem. For (b), fix a and r , and write $f = f_{a,r}$ and $I = \int f dx = k_a(r)/\sigma_a$. The realized weight is σ_a times the infinite trapezoidal sum $T_h = h \sum_{j \in \mathbb{Z}} f(x_j)$, $x_j = \ln u_{l_0} + jh$, minus two residuals. Let $u_j = e^{x_j}$.

Quadrature. By (H1) and [51, Thm. 5.1], $|T_h - I| \leq 2M/(e^{2\pi d/h} - 1)$ with $M = \sup_{|y| < d} \int |f(x + iy)| dx \leq L_a I$.

Lower bound on I . For $r \geq 2$, (H2) gives $I \geq c_- \int_0^{u^*} e^{-(r-1)u} u^{\beta_a-1} du \geq c_- \vartheta (r-1)^{-\beta_a}$.

Lower tail. The nodes $j < 0$ are replaced by one node at $u_0 = u_{l_0}$ with the same total weight. At lag r the residual is $\sum_{j < 0} h u_j m_a(u_j) (e^{-(r-1)u_j} - e^{-(r-1)u_0})$, which lies between 0 and $(r-1)u_0 \sum_{j < 0} h u_j m_a(u_j)$. By (H2) this is at most $(r-1)c_+ u_{l_0}^{1+\beta_a} \sum_{j \geq 1} h e^{-j\beta_a h} \leq (r-1)c_+ u_{l_0}^{1+\beta_a}/\beta_a$. Divided by the lower bound on I , it gives the second term of (27). It vanishes at $r = 1$.

Upper tail. The nodes $j \geq K$ are replaced by the delay mode, which is exact at $r = 1$. For $r \geq 2$ the residual is $\sum_{j \geq K} h u_j m_a(u_j) e^{-(r-1)u_j} \leq C_\infty e^{-(r-2)u_{hi}} \sum_{j \geq K} h u_j e^{-(1+\gamma)u_j}$ by (H3). The function $x \mapsto e^x e^{-(1+\gamma)e^x}$ decreases for $e^x \geq 1/(1+\gamma)$, so for $u_{hi} \geq 1$ the sum is at most $\int_{u_{hi}}^\infty e^{-(1+\gamma)u} du = e^{-(1+\gamma)u_{hi}}/(1+\gamma)$. Relative to I , this leaves the factor $(r-1)^{\beta_a} e^{-(r-2)u_{hi}}$. That factor is 1 at $r = 2$ and does not increase for $u_{hi} \geq \beta_+ \ln 2$, which gives the third term.

For (c), the choices of h , u_{l_0} and u_{hi} make the three terms at most $\varepsilon/2$, $\varepsilon/4$ and $\varepsilon/4$; the second uses $(R-1)u_{l_0} \leq 1$, so that the exponent $1 + \beta_a$ may be replaced by $1 + \beta_-$. A grid with $K = \lceil \ln(u_{hi}/u_{l_0})/h \rceil + 1$ nodes has a spacing at most h , and none of the three bounds increases when the spacing decreases. Finally, $1/h = O(\ln(1/\varepsilon))$ and $\ln(u_{hi}/u_{l_0}) = O(\ln R + \ln(1/\varepsilon))$, which gives (28). $\square$

Proof of Lemma 10. With $u = e^x$, $c = \cos y$ and $z = ue^{iy}$,

$$|f_{a,r}(x + iy)| = u e^{-(r-a)cu} |1 - e^{-z}|^a,$$

$$\int_{\mathbb{R}} |f_{a,r}(x + iy)| dx = \int_0^\infty e^{-(r-a)cu} |1 - e^{-z}|^a du,$$

and $I = \int_0^\infty e^{-(r-a)u} (1 - e^{-u})^a du$. For $\operatorname{Re} z > 0$, $1 - e^{-z}$ lies in the disc $|w - 1| < 1$, so $f_{a,r}$ is analytic in the strip. Two bounds hold for $|y| < \pi/2$:

- $|1 - e^{-z}| \leq \min(u, 2) \leq (1 - e^{-u})/\kappa_0$ with $\kappa_0 = (1 - e^{-2})/2$, because $(1 - e^{-u})/\min(u, 2)$ is smallest at $u = 2$;
- $|1 - e^{-z}| \geq 1 - e^{-cu} \geq c(1 - e^{-u})$, by concavity of $1 - e^{-u}$.

Hence $|1 - e^{-z}|^a \leq \kappa_0^{-a} (1 - e^{-u})^a$ for $a \geq 0$ and $\leq c^a (1 - e^{-u})^a$ for $a < 0$. Substituting $v = cu$ and using $1 - e^{-v} \leq 1 - e^{-v/c} \leq (1 - e^{-v})/c$ gives $\int_0^\infty e^{-(r-a)cu} (1 - e^{-u})^a du \leq c^{-1-\max(a,0)} I$. The ratio in (H1) is therefore at most $\kappa_0^{-a} c^{-1-a}$ for $a \geq 0$ and $c^{-1-|a|}$ for $a < 0$, for every r , which gives (29). The same bounds show that $|f_{a,r}|$ vanishes at both ends of the strip, since $1 + a > 0$ and $r - a > 0$. For (H2), $e^{-(1-a)u} \in [e^{-2}, 1]$ and $(1 - e^{-u})/u \in [1 - e^{-1}, 1]$ on $(0, 1]$. For (H3), $(1 - e^{-u})^a \leq (1 - e^{-1})^{-1}$ and $e^{-(1-a)u} \leq e^{-(1-\bar{a})u}$ on $[1, \infty)$. $\square$

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